

# **AxioVision**

**Perform to Perfection**

The Microscope Software for Industrial Applications  
– from Image Acquisition Through to Image Analysis,  
It's in a Dimension of Its Own.



We make it visible.

## **A New Way of Thinking**

Whether you are responsible for quality assurance in the aerospace industry or for the development of new materials, and whether you work in traditional fields such as materialography and geosciences or in the solar industry, these days digital microscope systems form the crucial basis for your applications. Carl Zeiss is driving this process with new solutions that are continually setting new standards. A major component is AxioVision, the microscope software from the microscope specialist. Thanks to its unique modular architecture, it is suitable for the novice and experts alike. Application-specific modules and tailored software packages offer attractive total solutions for defined tasks in science and industry. The AxioVision philosophy is uncompromising: the highest possible performance, easy operation, extreme flexibility, and seamless integration into the Carl Zeiss system world. A homogeneous solution.

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# An Easy Decision

Developed in close collaboration with users, AxioVision microscope software impresses through its practical relevance. Highly functional even in the entry-level version, it can be extended by modules available for sophisticated applications, thus satisfying in every detail.

## Easy start

AxioVision allows you to achieve outstanding results in digital microscopy and documentation. From image acquisition to processing, measuring, and annotating, to archiving and reporting, you can follow the process from beginning to end.

## Easy to use

AxioVision offers you a convincing operating concept: one that is simple and comprehensible, from the basic functions right through to the highly specialized analysis modules. *My AxioVision* allows you to adapt user interfaces and functions to your individual needs, configure your own toolbars and combine frequently recurring work steps in new dialogs. AxioVision transforms functional diversity and complexity into something quite simple for the user.

## Easy ZVI

ZVI is the name of the image format that stores your image data together with image number, acquisition date, microscope settings, exposure data, size and scale data, contrasting techniques used, etc. The advantages are obvious – the image information is available at any time. No annotations are lost and nothing is forgotten. A crucial point to note is that the annotations are not permanently burned into the image, but are stored in a file together with the image data. The image can be reproduced even years later under identical conditions.



### Easy economy

Offering the entire performance spectrum for contemporary digital microscopy at an outstanding price-performance ratio, AxioVision also excels from an economic standpoint. The fact that you can expand the system module by module in line with your own requirements means that you only

invest in the functions that you really need, whilst enjoying the security of being at the forefront of technological developments. With AxioVision LE, you even have a universal image viewer at your disposal free of charge for simple image analysis tasks.



# An Overview of All Modules



### Growing possibilities

The world of materials sciences is constantly changing and evolving, which means it requires a software package that can change and evolve with it. AxioVision is very flexible in its design. Because with every update and every expansion, Carl Zeiss is at the cutting edge of innovative software developments. In addition, the user interface is customizable. This gives users the ability to make it easy to understand and use for their specific applications. The functions of the basic program – imaging, processing, annotations, archiving, reporting, and microscope control – can be quickly expanded to meet your growing needs by adding further modules. Moreover, new solutions for specific applications are continually being developed. They include additional functions for image processing, interactive measuring, and automated image analysis as well as control modules for light path and motorized stages

#### Comparative Diagrams

Comparison of images with comparative diagrams

#### Topography

Height and roughness measurements

#### NMI

Measurement of non-metallic inclusions according to international standards

#### VBA

Integrated development environment

#### Particle Analyzer Projects

Measurement of particles

#### Asset Archive

Cataloguing and archiving of images, data sets, etc.

#### Commander

Recording/Automatic execution of AxioVision commands

#### Documentation

Image archiving and reporting

#### Configuration

Customization of user interface

*The decision for the basic AxioVision program is a sound investment in digital imaging. For a 100% compatible system solution that can be adapted at any time to your changing requirements and demands. A decision that not only protects your investment but guarantees you enormous flexibility.*

# Basic Program

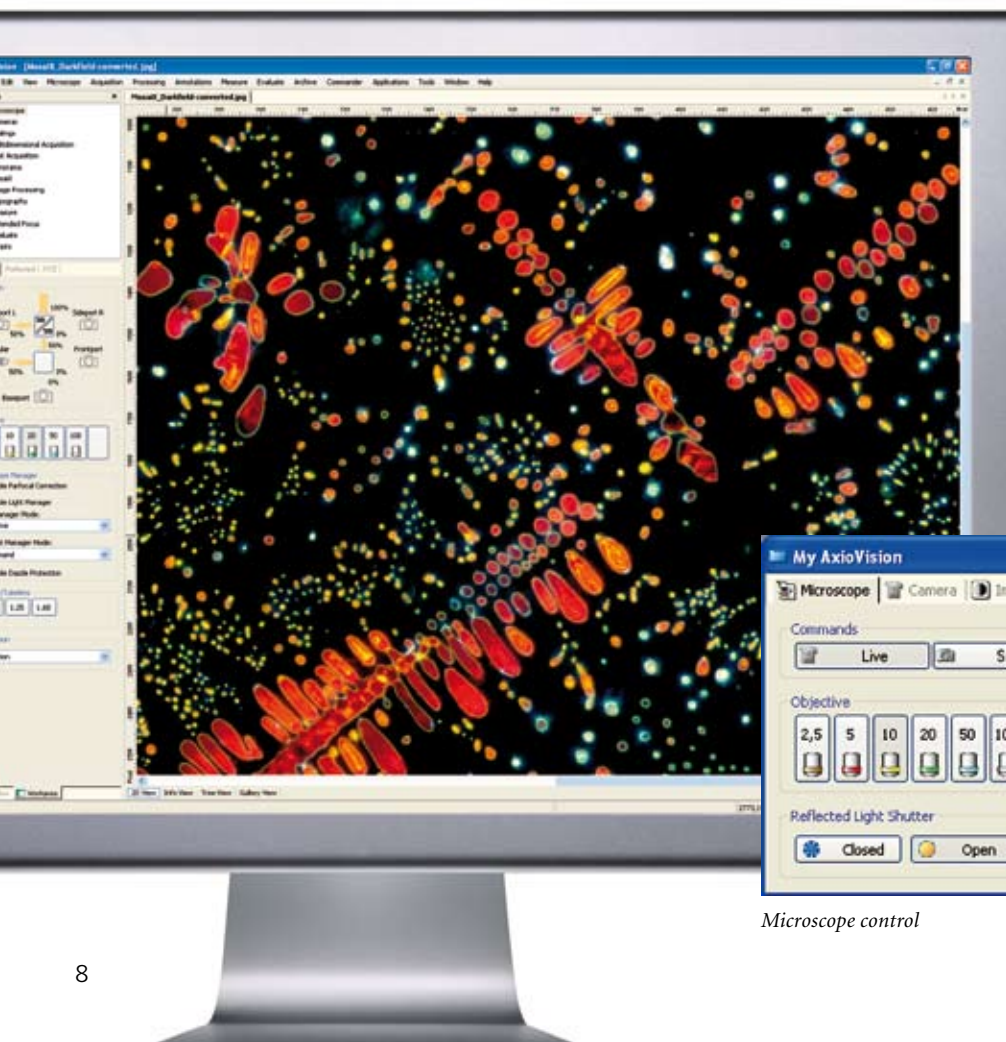
## Impressive Range of Functions

You will be amazed at the wealth of functions of the entry-level microscope software offered by Carl Zeiss. Even the basic version delivers a powerful image processing and analysis system, and meets all the key requirements of contemporary digital microscopy.

### Efficient microscope control

AxioVision allows you to control all motorized microscopes from Carl Zeiss – both automatically and interactively. Of course you can use manual standard microscopes as well. One of the advantages of software control is that you can

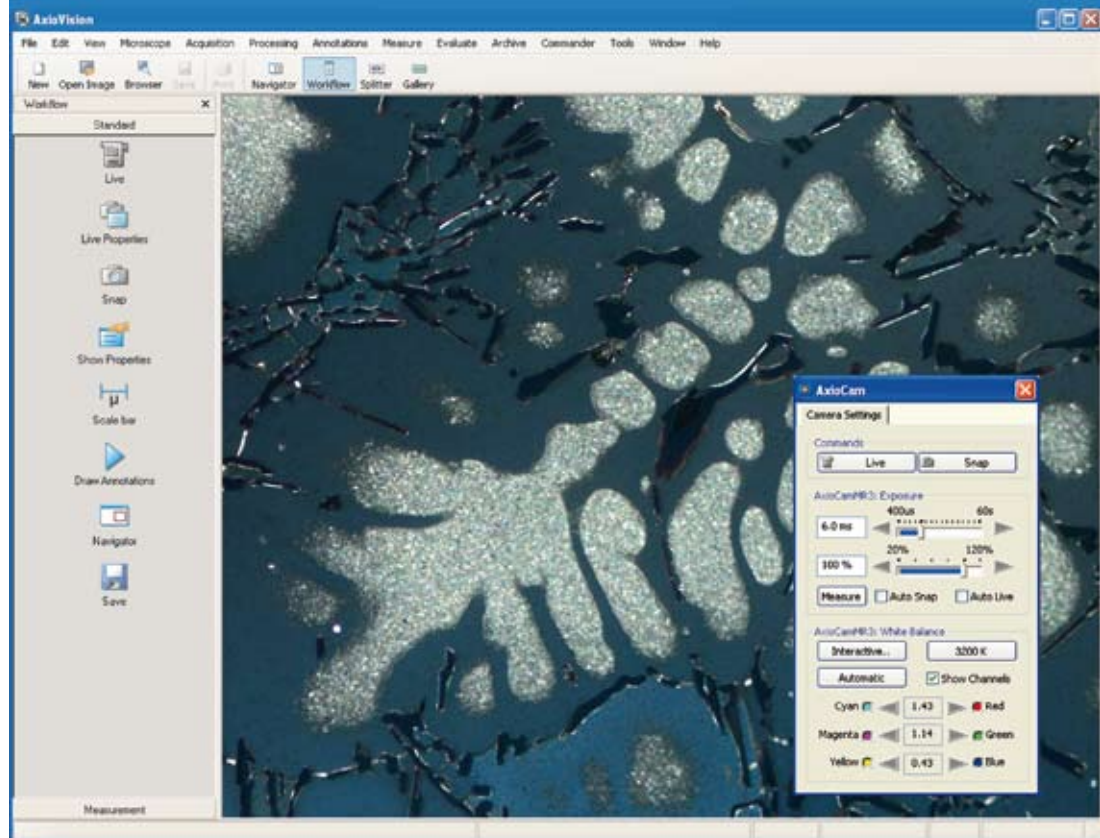
store desired microscope parameters quickly and easily, ensuring repeatability for subsequent analyses. In addition scaling factors and complex workflows can be recalled during analysis.



Microscope control



Digital camera



*Simple, clearly presented user interface for camera control and image acquisition*

### Flexible camera operation

Thanks to its interfaces for standard technologies, AxioVision allows you to use all types of cameras, from digital consumer cameras up to scientific microscope cameras. This includes the AxioCam family of cameras from Carl Zeiss. The seamless integration of cameras into the AxioVision software enables you to adopt all important image information and – with just a click of your mouse – to document your samples. The cameras from Carl Zeiss can also provide significant advantages in the areas of speed and resolution, automatic exposure settings and image acquisition. All cameras in the AxioCam family are controlled by the same operational elements.

### Rapid image processing

AxioVision offers you all the tools for:

- Contrast, brightness, and color control
- Noise suppression, smoothing, and contour enhancement
- Enhanced sharpness and detail emphasis
- Correction of illumination conditions and white balance control



*AxioCam*



*Image acquisition*



*Microscope (manual or motorized)*

## Basic Program

### Impressive Range of Functions

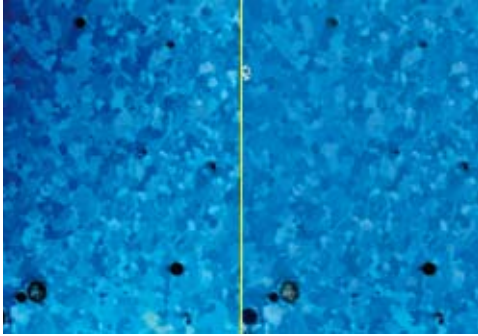
#### **Integration of text and graphic elements**

From scale bars and color markings to text and graphic elements – with AxioVision you can add all important annotations to your images using just one program. The corresponding scale is stored with each image, and scale bars can be automatically added at any time.

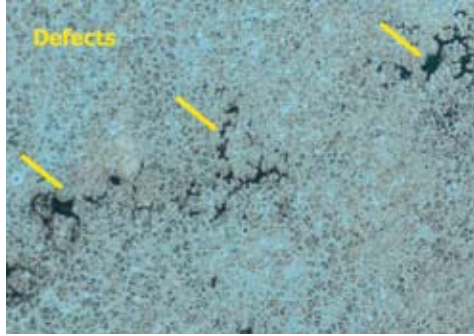
#### **Precise image measurement**

With the entry-level program, you can easily perform interactive measurements, such as length, area, and angles. The measurement data are available in a list, which can be easily exported to most spreadsheet programs, such as Microsoft® Excel.





*Image processing: correction of illumination error (left: with shading, right: corrected)*



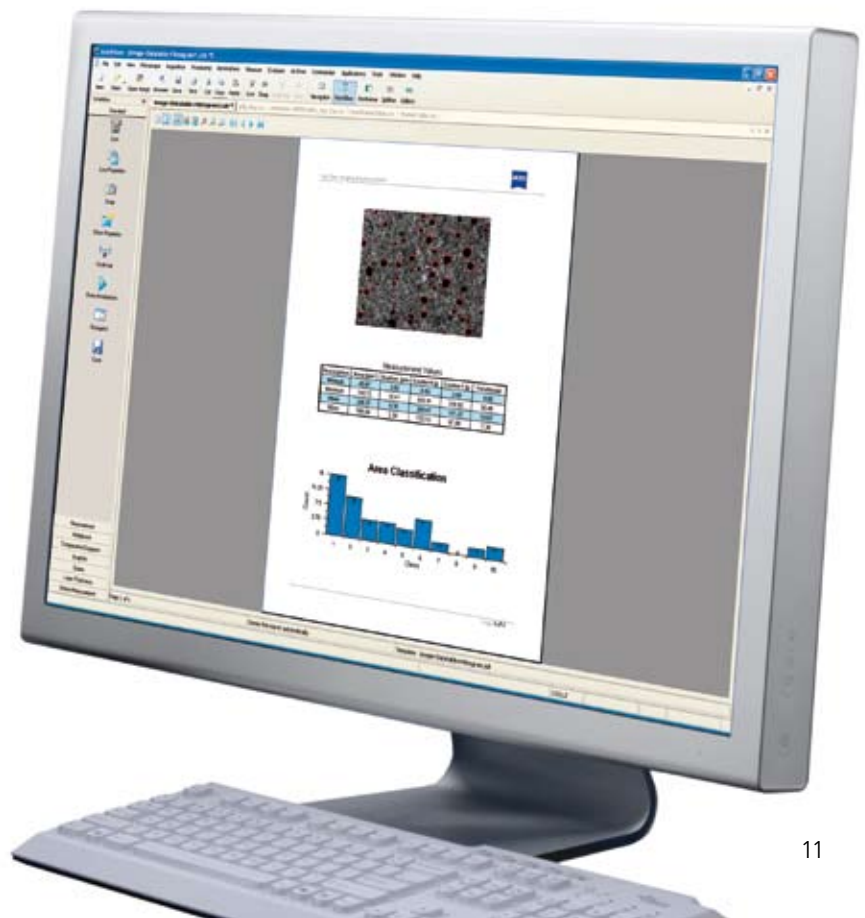
*Text and graphic elements: labeling of defects*



*Image measurement: area measurement*

### Perfect report generation

Whether using individually formatted or predefined layouts, AxioVision gives you all the options you need to generate effective reports or documentation. All information, such as measurement values, analysis results, tables, charts and images, can be conveniently arranged using predefined layouts or formatting defined by the user – simply at the touch of a button.



# Image Acquisition Modules

## Enhanced Performance in Imaging

The results of your analyses are only as good as the quality of your acquired images. AxioVision offers you the perfect basis for achieving the required quality with high-performance additional modules, from MosaiX and Autofocus through to Z-Stack and HDR Imaging. Modules that secure the additional information in your images that is so often crucial.

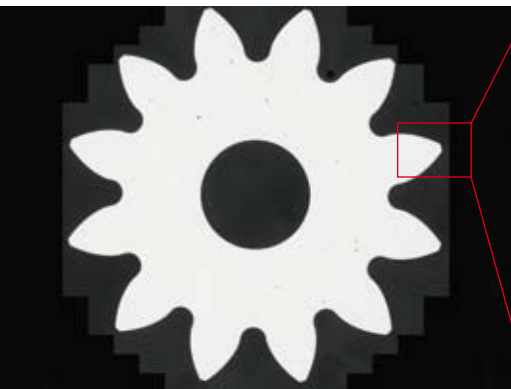
### Autofocus

The Autofocus module calculates the optimal focal position for a sample. The system is calibrated for each objective so that the software focuses accurately every time, or it automatically makes use of default parameters to achieve the correct focusing. In addition, with images that are acquired as a time lapse or at different positions, the system automatically refocuses. The Autofocus module works with all cameras that are directly controlled by AxioVision, provided that a microscope with motorized focus drive is used.

### MosaiX

Developed to acquire images of large surfaces, MosaiX scans your samples image by image and then combines these individual images to form a single MosaiX image. In

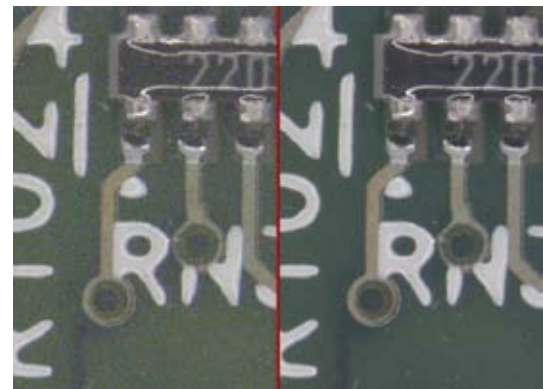
addition, uneven samples can be acquired without difficulty – thanks to the option of automatic focus adjustment. Prerequisites for this are a motorized focus drive and stage. Any loss in image quality caused by the overlapping of individual image tiles is avoided thanks to an intelligent mechanism that determines – either automatically or interactively – the position of individual image tiles on the basis of the image content, corrects their position and combines them to form a MosaiX image. The image you obtain preserves the high resolution of each individual image. As a result, it is not only suitable for navigating around the sample, but also forms the ideal basis for further analyses. Large objects, such as non-metallic inclusions, can be measured and the number of particles on a filter can be documented with equal ease. You are now no longer restricted by the limits of an individual image.



*MosaiX image of a gear-wheel, 15x20 tiles*



*Detailed view: individual tile of the MosaiX image*



*Images of a printed circuit board – on the left, a single image and, on the right, an HDR image*

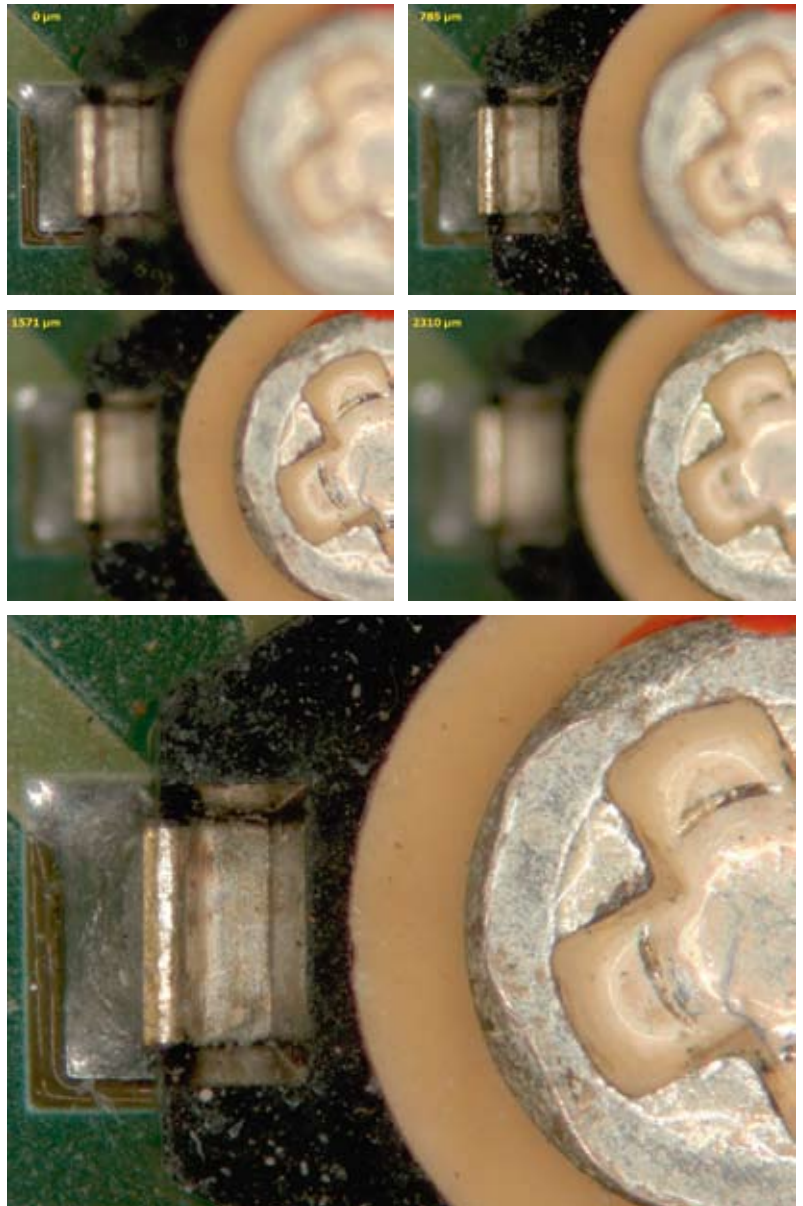
### HDR Imaging

Developed to enable you – when using reflected light microscopy, among other techniques – to capture all object details on specimens with very high contrast in a single high-resolution image: the HDR Imaging – High Dynamic Range Imaging – module expands the dynamic range of your cameras, increasing the brightness gradations in your images measurably. Up to 32 images with different exposure times are acquired one after the other and merged to produce a single resulting image. This means reduced noise and improved image quality. Depending on the specimen and the acquisition situation, you will obtain significantly more information and, consequently, more precise measurement results. This technique is also an inexpensive alternative if you want to improve the image quality of your camera, as even in the case of a camera with a low dynamic range the results are clear to see. It is an economical way to transform an 8-bit camera into a camera with a 12-bit dynamic range, for example.

### Extended Focus

A microscope's depth of field is often not sufficient to obtain a single image which is sharp over the whole field. The software solution to this problem is the Extended Focus module. The principle is simple – while focusing through the sample, you acquire a number of images at different focus positions or use your Z-stack images as input data. In both cases the sharp details from each individual image are extracted and a final image is calculated on the basis of state-of-the-art algorithms. The result is an image of first-class quality that is rich and sharp in every detail.

*Single images from different focus planes of an electronic component.*  
*With Extended Focus, users can achieve an image that is sharp across the whole field.*



# Image Acquisition Modules

## Enhanced Performance in Imaging



*Panorama: a crack is followed beyond the image field. Only the required images are acquired.*

### Z-Stack

To enable the automatic generation of Z-stack images, the software controls the z-drive of a motorized microscope in precise steps. This is always synchronized with acquisition. You can either determine the focusing interval yourself or have it automatically computed for highest sample accuracy. The advantage of this module is the optimal detection of information in the third dimension. In addition, with the Cut View function, even the entry-level version of AxioVision provides you with a highly effective technique for Z-stack analysis.

### Panorama

Perfect for objects that do not fit into a single image field – the Panorama module is the manual variant of MosaiX. With the help of a manual mechanical stage or coded stage you can generate high-resolution panorama or overview images from individually acquired images, making it possible to follow irregular structures, such as cracks, beyond the edges of the frame. Overlapping images can also be combined with pixel precision, ensuring that the important details of your sample are all contained in a single image.

### Mark&Find

This module is used to record, store, and automatically revisit different positions on your samples. It requires the use of motorized x-, y-stages. The positions on the sample are stored together with the acquired image. Lists of positions can also be imported. Your advantage: reliability, time saving, and statistical accuracy.

### Time Lapse

Investigating changes over time, documenting results clearly – with the Time Lapse module, you can precisely control both camera and microscope. With the help of a heating stage it is possible to monitor structural changes, for example.

### Combinations

MosaiX, Autofocus, Z-Stack, Mark&Find, Time Lapse – all these modules can be freely combined with each other, creating solutions capable of precisely meeting a wide range of demands. The result is the cost-effective adaptation of individual solutions to a specific application – with no unnecessary investments.

# Image Processing Module

## Get More from Your Images

All the important digital image processing techniques in a single module – Imaging Plus allows you to process your images for maximum information content and the best analysis results.

### Imaging Plus

- **Image Enhancement**

In addition to improving contrast, brightness and color, this function compensates lighting deficiencies and shading. Filters for smoothing, sharpening, and edge detection are included as well as user-definable filter operators.

- **Gray Morphology**

An ideal tool for grain boundary reconstruction, for example, enabling you to reconstruct the boundaries of joined objects exactly. The advantage: individual objects can be separated with precision.

- **Image Arithmetics**

The process of calculating a new image from existing images pixel-by-pixel: AxioVision Imaging Plus allows the quantitative combination and comparison of images.

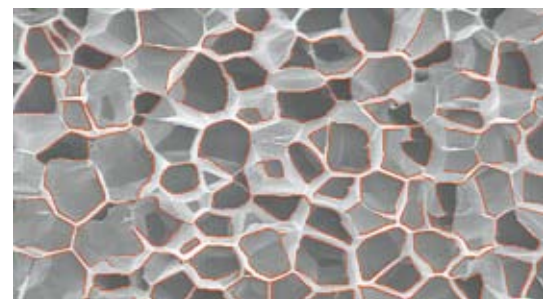
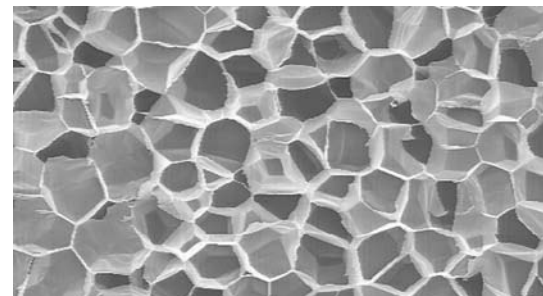
- **Elastic Registration**

The solution for achieving congruence between two images with the same content that cannot be corrected simply by shifting, rotating or adjusting the size of the images.



*By enhancing the edges, object boundaries can be detected with greater precision.*

*Morphology functions permit the exact reconstruction of object borders, thus preparing them for automatic measurement.*



## Image Analysis Modules

### Uncompromising Precision

Utilizing all the information of an image: AxioVision offers you a powerful spectrum of additional modules for image analysis. For greatly simplified processes, faster results, uncompromising reliability and maximum reproducibility.

## Interactive Measurement

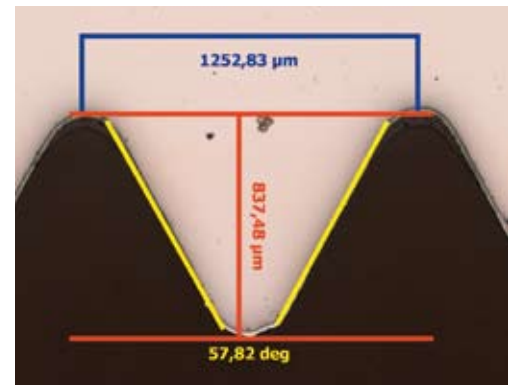
With this module, parameters describing the specimen can be determined interactively (e.g. size). A measurement program wizard allows users to exactly determine which measurements should be taken. All parameters are then executed in the specified order. As a result, geometric and densitometric parameters are presented in a straightforward measurement list, to be stored with the image in the archive. You can retrieve this information later at any time. In addition, all requested measurement values can be exported (e.g. into Microsoft® Excel).

## Online Measurement

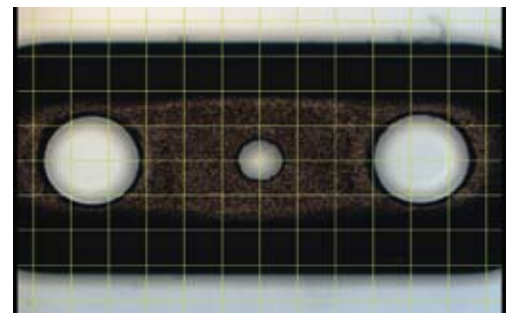
Measure samples direct on the monitor, without the need to acquire images. With this module you can analyze structures interactively, directly in online images, meaning that visual inspections can now be carried out quickly and conveniently on screen. All the measurement tools that you employ for your acquired images can also be used here. You select the desired parameters from a choice of up to 90 options.

## AutoMeasure

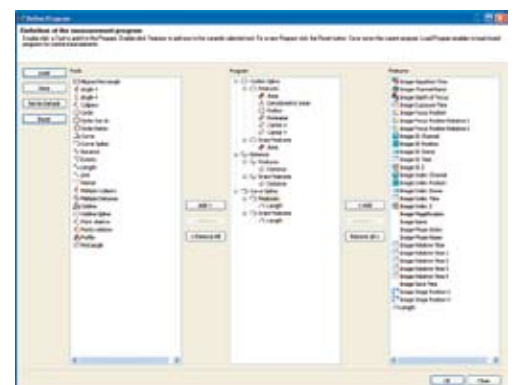
If you need to create automatic measuring routines yourself: with the AutoMeasure module you can rapidly obtain precise results –without any complicated programming. With the help of a measurement wizard, AutoMeasure enables you to carry out complicated measurements within a few minutes. Simply define the programs that you need and you can measure an unlimited number of images – while completely controlling the measuring process. You can determine which steps to be conducted. Even automated processes can be interrupted at any time and all parameters individually adjusted with the function dialog.



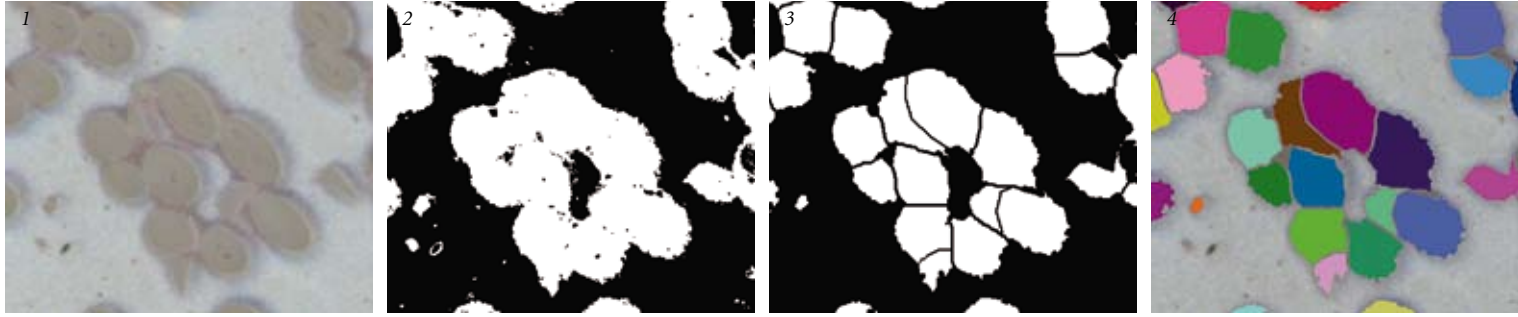
### Interactive measurement of distances and angles



### Assessment of hole separation in the online image



### Measurement parameters – list of options



Automatic analysis of a polymer sample (fig. 1): threshold segmentation (fig. 2), correction of the binary image and automatic separation of objects (fig. 3), overlaid display of the measurement in the resulting image (fig. 4)

### AutoMeasure Plus

Capturing the entire structure of the image automatically – now possible in a single measurement step with this module. The result: fast, precise and reproducible quantitative analyses. Further advantages: the direct access to all functions via the menu and the option to combine with the automatic processing module Commander. It enables you to merge the results of repetitive work steps in a single command – ideal for the automatic processing and reproduction of standard lab assignments. This module consists of three functionality groups:

- **AutoMeasure Plus – Segmentation**

This function offers threshold operators for monochrome and color images that are necessary to identify your objects. The objects can also be identified with the click of a mouse using "Region Growing". These two methods are supplemented by complex methods for segmentation, including dynamic and automatically generated threshold values as well as edge detection. The result is a binary image in which all specimen pixels are white and all background pixels black.

- **AutoMeasure Plus – Binary image processing**

Functions for linking, masking, and filling holes ensure that the binary image is optimally prepared for measurement. Artifacts are removed and contours smoothed.

- **AutoMeasure Plus – Automatic measurement**

This function makes it possible to determine morphometric measurement parameters from the contour of the specimen. The binary image is used as a mask to calculate geometric and densitometric parameters from the original image. Results can be imported into Microsoft® Excel – ideal for generating statistical information about specimen details.

### TIC Measurement

The TIC module (TIC – **T**otal **I**nterference **C**ontrast – only in combination with the TIC slider from Carl Zeiss) allows precise, contact-free and, therefore, extremely simple determination of the optical height and thickness of object structures over a range from just a few nanometers through to several micrometers. The advantage of the TIC interferometric method lies in the combination of short measurement and analysis times with a high degree of accuracy. The use of circularly polarized light makes the orientation of the structures on the sample irrelevant, eliminating the need for stage rotation. It is even possible to analyze samples with large surface areas.

# Topography Module

## Surface Analysis Using Digital Height Data

Whether you are dealing with height measurements for production control in the automotive industry, roughness measurements or 3D reconstructions for the optical assessment of production tolerances – the Topography module generates height maps of your samples automatically or interactively, using a contact-free and non-destructive method.

### Functionality

Your work with this module is based on Z-stack images of your samples, interactively acquired images from various focus planes or stereo image pairs acquired via both ports of a stereomicroscope. From these, the Topography module generates the following for you:

- 3D topographies from various perspectives, with texture, grid or surface shading
- Roughness measurements in accordance with EN ISO 4287
- Height measurements

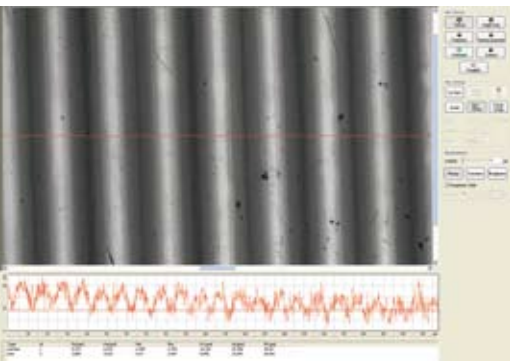
The measurements are performed along a profile line that you have drawn into the topographic image using a measurement tool. Several profiles can be drawn in and simultaneously selected. The measurement results are displayed in a table. The parameters of the primary profile, waviness or roughness are displayed on the basis of the cut-off

wavelength ( $\lambda$ ) selected. A wide range of functions are also offered in the module, from the display of sharp regions as a texture image and dynamic flooding in the height image through to the possibility of modifying projections and height images via look-up tables. All height information is stored in the image and is available at any time.

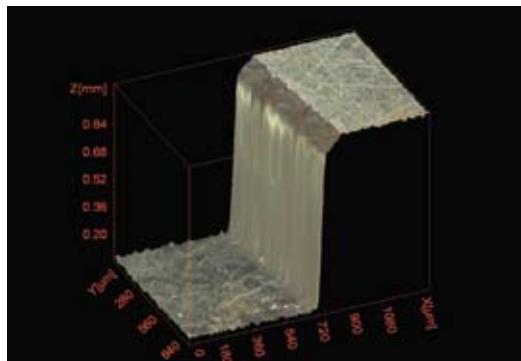
### Integrated guidelines

The AxioVision Topography module calculates the following roughness parameters in accordance with EN ISO 4287:

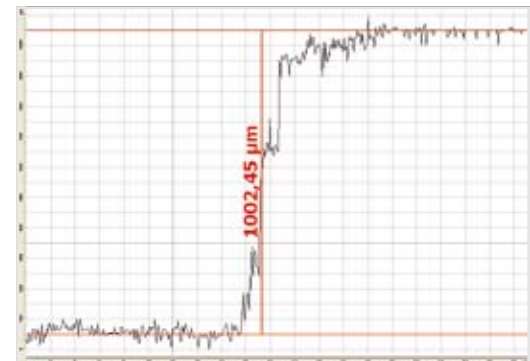
- $R_a$ : arithmetic mean of the profile ordinates
- $R_q$ : quadratic mean of the profile ordinates
- $R_{sk}$ : skewness of the profile
- $R_{ku}$ : steepness of the profile
- $R_v$ : depth of the greatest profile valley
- $R_p$ : height of the greatest profile peak
- $R_t$ : total height of the profile



Roughness measurement in accordance with EN ISO 4287 along a user-selected profile



3D topography (texture projection) of a gauge block



Height profile measurement in a topographic image

# Archiving Module

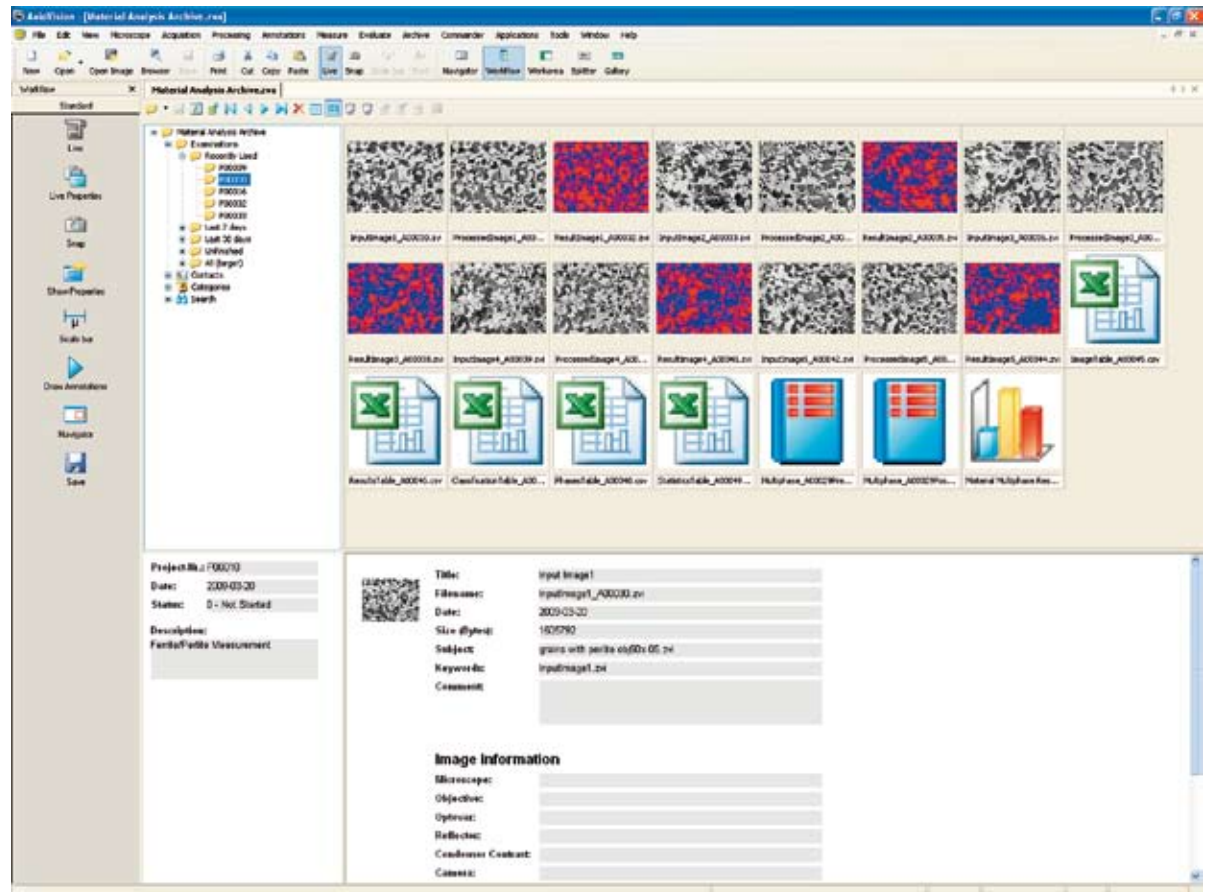
## Well-conceived Data Management

Maintain an overview of your images, measurement results, and reports – AxioVision allows you to manage all your data simply, transparently, and completely.

### Asset Archive

The powerful AxioVision Asset Archive module allows you to archive not just your images, but also all the associated image data, measurement results and reports of your analyses – in an extremely simple way under a single project number. This makes the process of finding your way around a large number of data sets significantly easier and quicker. The up-to-date image management software offers a range of benefits:

- Fast, flexible search functions: search by projects carried out for a certain customer/client, by projects carried out in the last week/month, by image or sample name, by date, labels, etc.
- Clear display of all key data acquired with the image
- Logically organized, hierarchical structure:  
Customer/client → Project/assignment → Asset
- Management of customer data/contacts/projects



Asset Archive: structured storage of related images, measurement results and reports in one project

# Configuration Modules

## Automatically Faster

With *My AxioVision*, even the entry-level version of AxioVision offers you all kinds of methods for creating individual operating windows. Possibilities that, with the help of two additional modules, can be expanded almost infinitely – right through to the development of your own programs within AxioVision.



*User-defined dialog for operating the microscope and camera*

### My AxioVision

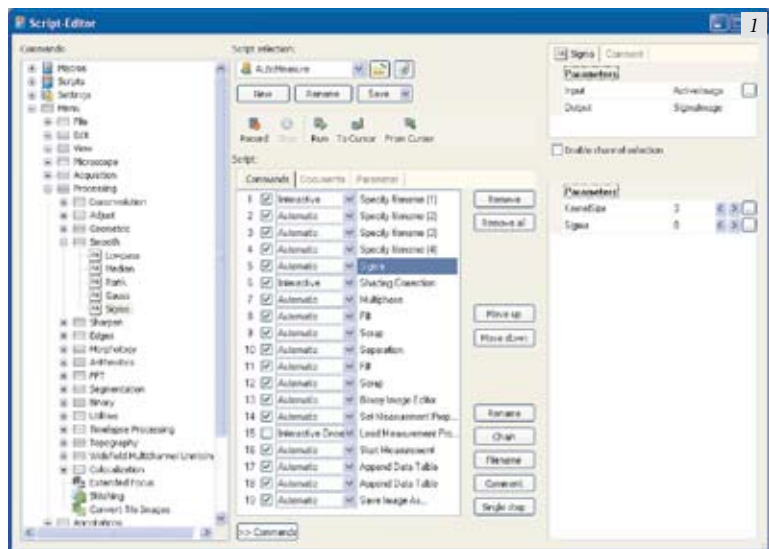
*My AxioVision* is the design concept behind the AxioVision architecture. It forms the basis for the almost unlimited freedom that AxioVision gives you when it comes to adapting the powerful microscope software to your own particular requirements or for several different users. AxioVision stands out thanks to the unique range of options it offers for designing customized user interfaces and functions. Perfect for simplifying, optimizing and increasing the efficiency of your digital microscopy using comprehensible workflows. Even in the entry-level version the Carl Zeiss software offers you plenty of scope to structure your individual work environment: you can configure your own toolbars and combine relevant elements for camera and microscope control in your own new dialogs. Elements can be removed or added – depending on what you require for your work process.

### Commander

The Commander module allows you to record subsequent steps of your workflow, edit and refine these steps, set parameters, and make all this available under a single command. The benefits are impressive: automatic processing of typical lab assignments and complete reproducibility of the results, in addition to fast adaptation to new requirements.

## VBA

Perhaps you need more functions than the wide range that AxioVision provides. In this case, it is possible to increase and extend the performance of Carl Zeiss software, and adapt it to your needs, with VBA (Visual Basic® for Applications), the programming language Carl Zeiss uses for AxioVision functions. VBA provides a completely integrated development environment that is familiar to programmers. Since VBA is directly integrated into the host application, it offers the advantages of fast internal cooperation as well as the opportunity to develop solutions without additional programs. The results look and act just like AxioVision. The big advantage of this module is that a minimum of training time is required for the users of your individually developed software.



- 1) Commander window for recording work steps for automatic procedures
- 2) User-defined toolbar with daily workflow

# Materials Packages

## Highly Versatile

Compact and economical: with the materials packages, AxioVision offers you complete solutions that can be put into practice immediately for your routine tasks in industry – at an all-inclusive price. They can be used with any microscope, are easily upgraded and come fully preconfigured – from image acquisition through to analysis, evaluation and archiving software.

### Three packages for three different performance requirements

The materials packages combine the strengths and performance of important AxioVision modules, starting with the functions of the basic program through to modules such as AutoMeasure Plus or MosaicX. Three tailored packages are available:

- Materials Core: for routine applications
- Image Analysis upgrade: for advanced measurement tasks
- Motorization upgrade: for automatic acquisition using motorized microscopes and stages

And, as you would expect from Carl Zeiss, it is possible to upgrade these packages easily and economically to gain access to the next performance class. You can therefore be confident that, even with the compact solution, you will be able to respond flexibly to new performance requirements at any time.

### What the materials packages offer

|                        | Image Acquisition | Extended Focus | Panorama | Autofocus | Z-Stack | MosaicX | Image Processing | Imaging Plus | Image Analysis | Interactive Measurement | Online Measurement | AutoMeasure | AutoMeasure Plus | Documentation | Asset Archive | Configuration | My AxioVision |
|------------------------|-------------------|----------------|----------|-----------|---------|---------|------------------|--------------|----------------|-------------------------|--------------------|-------------|------------------|---------------|---------------|---------------|---------------|
| Materials Core         | X                 | X              | X        |           |         |         | X                |              | X              | X                       | X                  | X           |                  | X             | X             | X             | X             |
| Image Analysis upgrade | X                 | X              | X        |           |         |         | X                | X            | X              | X                       | X                  | X           | X                | X             | X             | X             | X             |
| Motorization upgrade   | X                 | X              | X        | X         | X       | X       | X                |              | X              | X                       | X                  | X           |                  | X             | X             | X             | X             |

 AxioVision Basic program  AxioVision module

# Application Modules

## Greater Efficiency in All Routine Tasks

Whether for particle analysis or grain size analysis – the AxioVision application modules offer complete application solutions for industrial practice. They have been developed to allow reproducible results to be achieved with as little interaction as possible. As well as a transparent system design that is identical for virtually all application modules, this is realized using fully automated processes – making your routine tasks faster, more efficient and more reliable.

### Operating modes

The AxioVision application modules have been systematically developed for use in practice and are focused on two key requirements: flexibility and security. The flexibility of being able to add in-house test specifications and new standards independently at any time – without external assistance or additional programming. And the security of knowing that there is no possibility of careless changes to settings leading to incorrect measurement results. The operating concept of all application modules is therefore based on two functional areas: all key settings for the measurement routines are defined in administrator mode. Measurements are then carried out in user mode – in a simple way via a transparent user

interface. The administrator will have defined in advance whether and to what extent users can change measurement parameters here. System operation is incredibly easy and has been designed to allow reproducible results to be achieved with as little interaction as possible. The performance of a measurement can be automated to such an extent that only the project data need to be entered – the entire analysis process runs automatically.



# Particle Analyzer System

## Particle Analysis with No Limits

Nowadays, guidelines and standards place numerous sectors under an obligation to guarantee and document the cleanliness of their products in a verifiable way. For example, the pharmaceutical industry is subject to FDA Regulation 21 CFR Part 11 and the automotive industry to the standard VDA Volume 19. These require the analysis of complete samples and the standard-compliant acquisition of images of even the smallest particles.

### Functionality

Analyzing residual dirt on filter membranes, examining lubricants, identifying defects in die cast components, inspecting distributions of active substances in crystalline form in ointments, detecting, separating and measuring plaques in microtiter plates – behind the AxioVision module Particle Analyzer Projects lies an integrated total system for a versatile range of applications. Tailored to industrial practice, this system impresses with its performance and ease of operation. It automatically sets classes and boundaries and selects the measurement parameters. The particles can be classified on the basis of various types (reflective, non-reflective, fibers, etc.). Measurement results are presented clearly in an image gallery and can be edited. It is also possible to relocate, further process, delete or reclassify particles – an important function that enables you to control and correct your measurement results at any time. The key to the system's convenience and operational reliability is the fact that, on the basis of the motorized microscope platform, the entire

measurement procedure – from image acquisition and particle detection through to analysis and report generation – is performed fully automatically. For microscopy you can reliably reproduce.

### Integrated guidelines

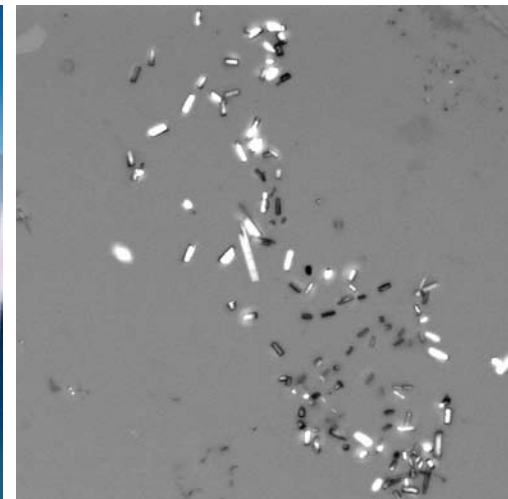
The Particle Analyzer system supports all key national and international standards: e.g. ISO 4406, ISO 16232 and VDA Volume 19. The corresponding classes and class boundaries are set automatically and the relevant measurement parameters selected. Within the context of GxP projects, as an option Carl Zeiss can also supply – with AxioVision GxP\* – the necessary software and documentation. A certifiable calibration standard is also available as an option for system inspection. The analysis system is configurable and you can easily add new standards to the existing system yourself, without having to wait for new software.

\* Module not available in all countries

*The system variants of the Particle Analyzer, based on SteREO Discovery.V12 and Axio Imager.Z2m*



*Active substances in crystalline form in ointments*



# NMI System

## Determining Non-metallic Inclusions in Steel

Current and new industry standards define a wide range of requirements for determining the content of non-metallic inclusions in steel. NMI is integrated into a total system with a practical orientation and enables you to determine the content of non-metallic inclusions – reliably and with absolute precision in accordance with established steel standards.

### Functionality

Developed in collaboration with experienced users involved in the production and application of steel, NMI offers a powerful analysis method that allows you to respond reliably to and comply with the requirements of even the latest industry standards. With NMI you control all system components fully automatically via the software. Workflows are automated with the help of test specifications, which can be adapted individually. Once it has been set up, the system scans the surfaces of your steel samples independently and analyzes them. Inclusions are captured in full even if they are larger than the field of view of your camera.

### Integrated standards

NMI analyzes your steel samples simultaneously in accordance with the guidelines of various standards:

- EN 10247
- DIN 50602
- ASTM E45
- ISO 4967
- JIS G 0555

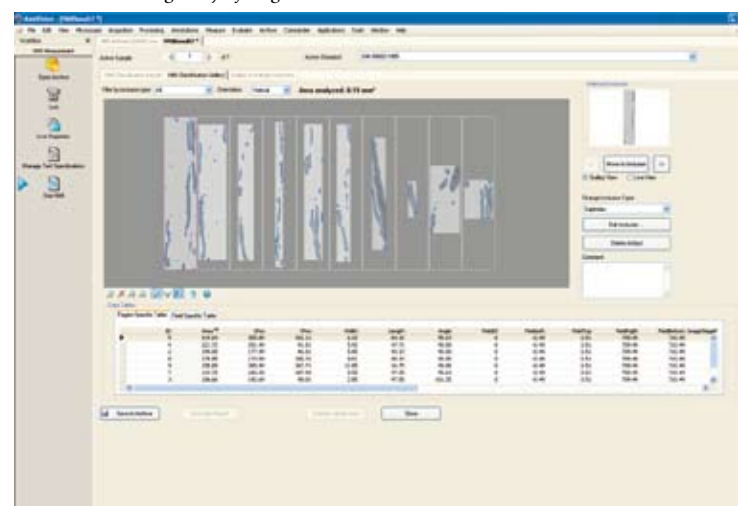
### Results display and data management

The results display in NMI sets the standard when it comes to transparency and depth of information. A wide range of options are available for displaying the measurement values, from the results view in image and tabular form through to the selection of various gallery views offering all measurement and classification data, elimination of artifacts, etc. Reports are generated automatically according to the requirements of the selected standard and can be modified individually. All measurement data, such as tables, images and reports, but also the test specifications, are saved and managed in an Asset Archive. The archive features search and filter functions, ensuring that the data can be easily found and called up again at any time.

*The system variants of NMI, based on Axio Imager.Z2m and Axio Observer.Z1m*



*Results view with gallery of largest inclusions*



# Grains

## On the Trail of Structures

A standard-compliant and flexible solution for analyzing grain sizes in materialographic samples – Grains offers you an easy-to-use analysis tool for determining grain sizes in accordance with the requirements of international standards. Fully automatically or interactively and with high precision and reproducibility.

### Functionality

Three measurement modes are available for analyzing grain sizes: Comparison, a purely interactive method for comparing micrographs with comparative diagrams; the semi-automatic intercept method; and the automatic method, which automatically reconstructs grain boundaries for you and determines the individual grain size. You decide which method is appropriate with a click of your mouse, depending on the characteristics of the sample.

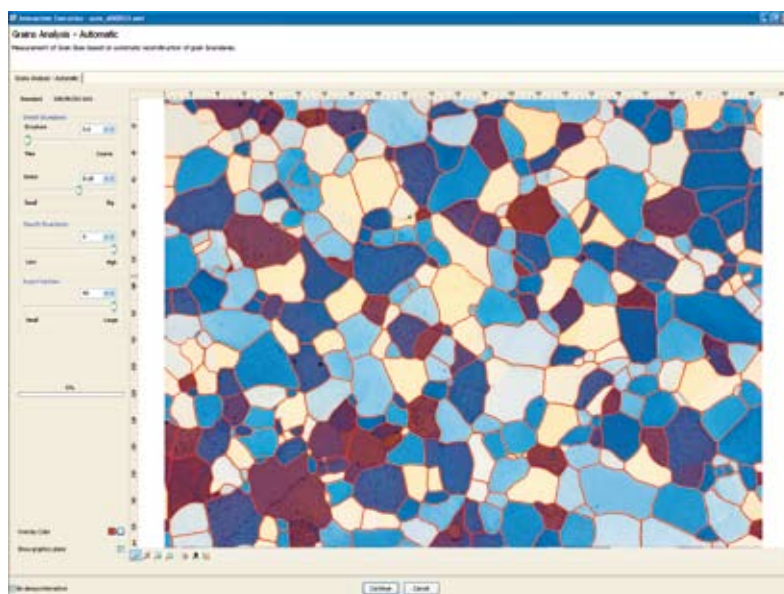
### Integrated standards

Grains analyzes your samples in accordance with the guidelines of the following standards:

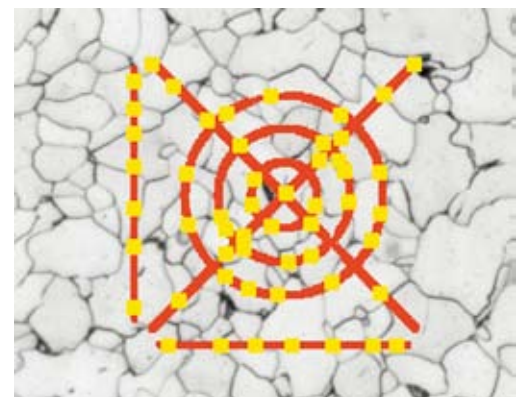
- ASTM E 112
- ASTM E 1382
- DIN EN ISO 643

Additional standards for the Compare method:

- ASTM – Plate I, Plate II, Plate III, Plate IV
- SEP 151061
- BS 4990



Automatic grain size analysis



Resulting image for intercept method

# Multiphase

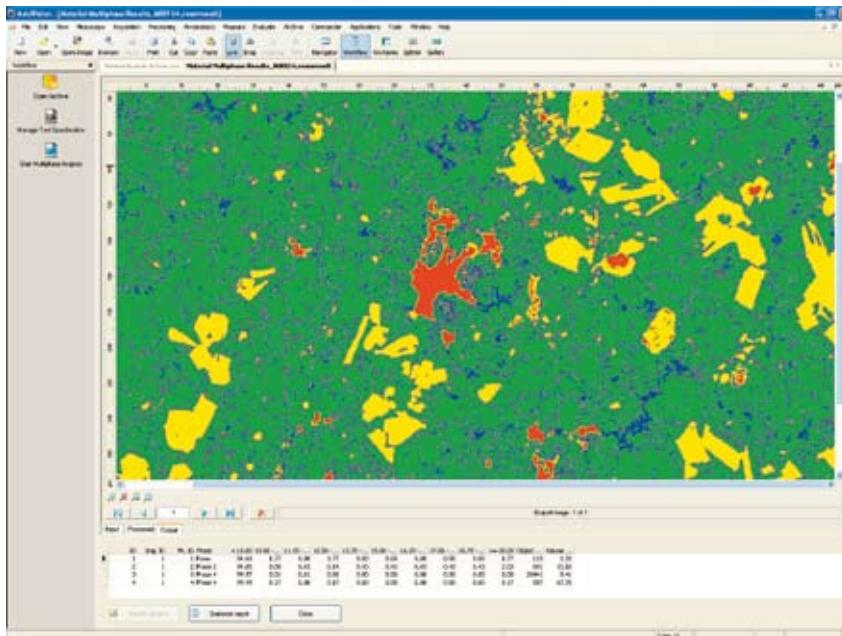
## Greater Flexibility for Phase Analyses

Multiphase analyzes the phase distribution in your samples for you. Phases are precisely measured on the basis of parameters such as size, shape or orientation and documented clearly in terms of the percentage of the area they represent, as classified particle sizes or in the form of a comparison – quickly, precisely and reliably.

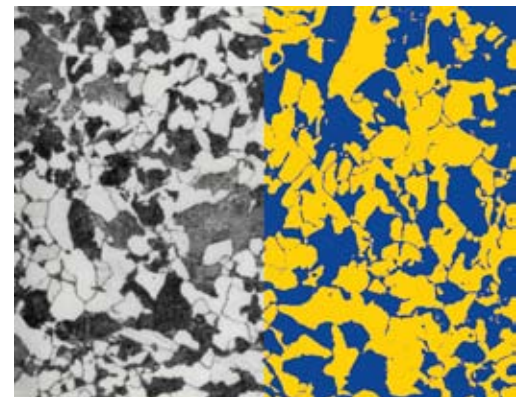
### Functionality

With Multiphase you decide whether you want to determine the percentage of the area that the selected phases represent or the size of the individual particles of a phase. The measurement values are then classified to provide you with a better overview. Complete flexibility is offered thanks to the free definition of parameters: individual class boundaries, linear or logarithmic classification, weighting on the basis of area or number. The parameters for the measurement can be drawn freely from the complete range of AxioVision measurement functions, and even user-defined parameters

are possible. If the standard steps prove insufficient, experienced users have the option of tailoring the image-processing steps to meet their individual requirements. The results are then displayed in a clearly structured report – the standard reports supplied can be modified individually.



Multiphase analysis: results view



Ferrite and perlite

# Layer Thickness Measurement

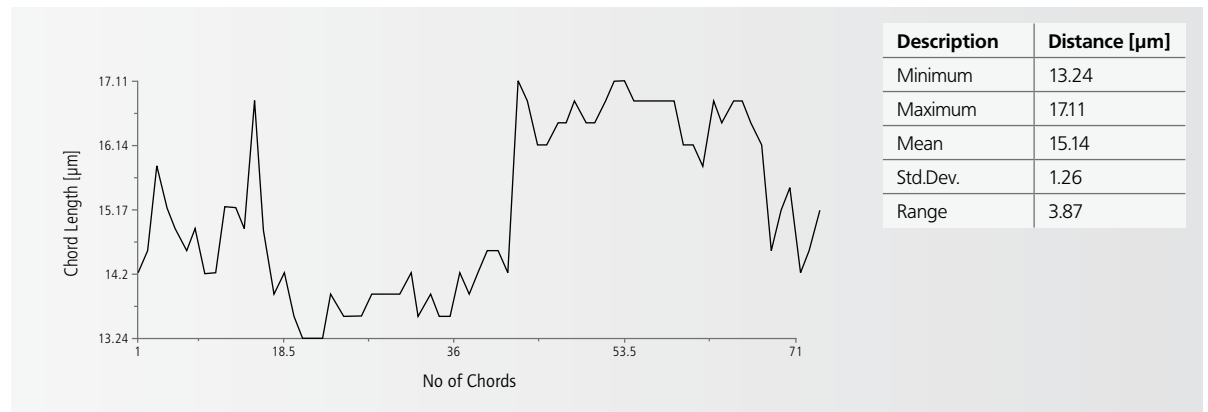
## Precision in All Layers

Convenient and economical: Layer Thickness Measurement is a powerful tool for determining layer thicknesses in your samples – from simple to complex layers and from individual layers through to several layers.

### Functionality

Layer Thickness, which can be used universally to meet your specific requirements, offers you a choice of different methods for detecting the individual layers of your samples. In the first step, the layers are identified on the basis of the color or gray value or drawn in interactively. The module then automatically calculates the course of the measurement chords, depending on the layer gradient present. It does this precisely and individually for each layer and independently of the number of layers. How the measurement will be performed is always your decision. You choose the number of

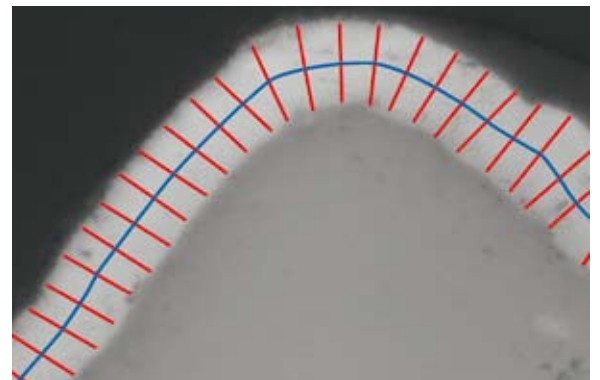
measurement chords or the distance between the measurement chords and, therefore, determine the accuracy of the result. This takes the form of a clearly structured report containing sample data and measurement values, such as the maximum and minimum chord lengths, mean, and standard deviation. In addition, a distribution of the chord lengths can also be displayed graphically. The fact that, if you so wish, you can also use all the other statistical analyses available in AxioVision is another important advantage.



Cross-section polish of a coated screw



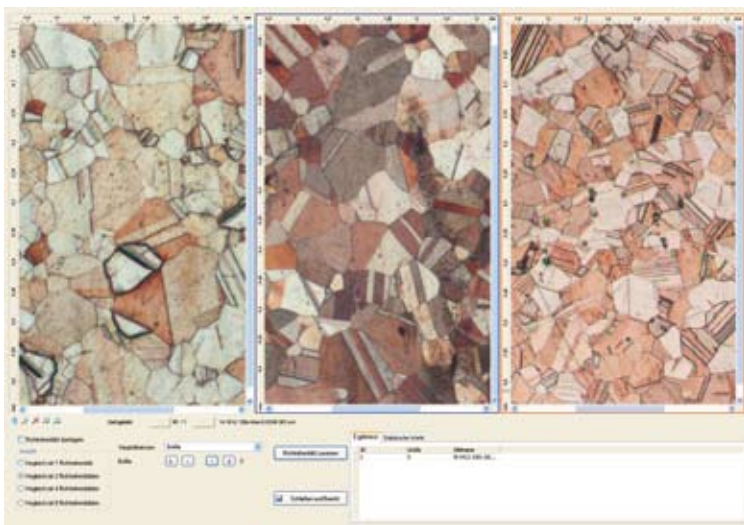
Coating with measuring chords



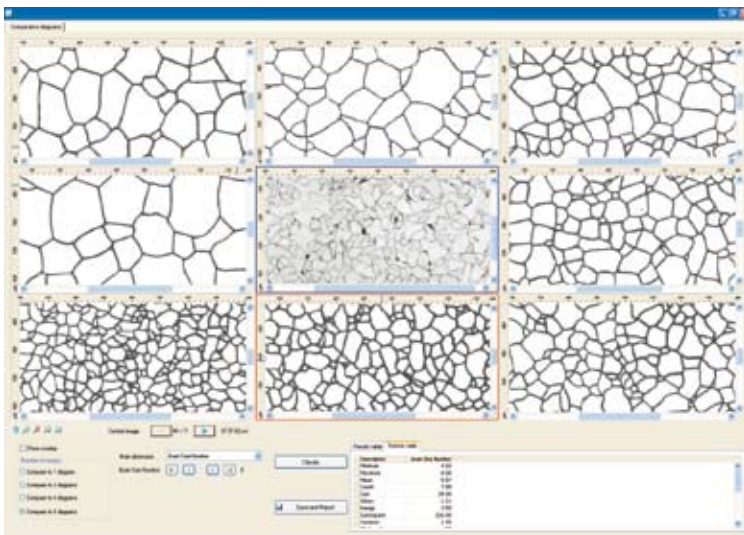
# Comparative Diagrams

## Assessing Structural Parameters

Assessing structural parameters conveniently on screen, generating comparative diagrams yourself – Comparative Diagrams brings a new level of quality to the interactive comparison of defined parameters and replaces the previous method involving the use of comparative diagram charts. It is a powerful tool for metallographers, materials scientists and quality assurance professionals.



User-generated: the image of the sample is flanked by two comparative diagrams



Comparative view with eight comparative diagrams

### Functionality

Comparative Diagrams, which serve as an aid for the comparison of samples under the microscope, are collections of images that have been assigned certain characteristics – structural characteristics, for example, or good/bad preparation. Comparative Diagrams displays the micrographs of the samples you wish to examine together with the comparative diagrams on screen and, therefore, makes interactive comparison possible. The result is a table containing the comparative diagram numbers relating to each image and a table containing statistical analyses. You can also have a micrograph displayed with an overlaid comparative diagram.

### Integrated standards

It is possible to generate your own comparative diagrams, with no additional costs and no waiting time: using the wizard function you can adapt the comparative diagrams simply to your own requirements. The following standards are included in the software package:

- EN ISO 643
- ASTM E 112
- ASTM E 1382
- BS 4990
- SEP 151061
- EN 10247
- DIN 50602

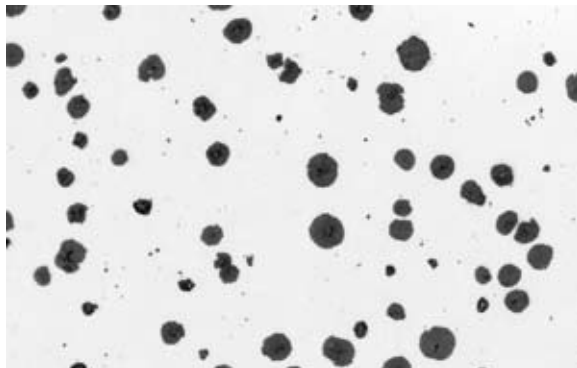
# Graphite

## Classifying Graphite Particles

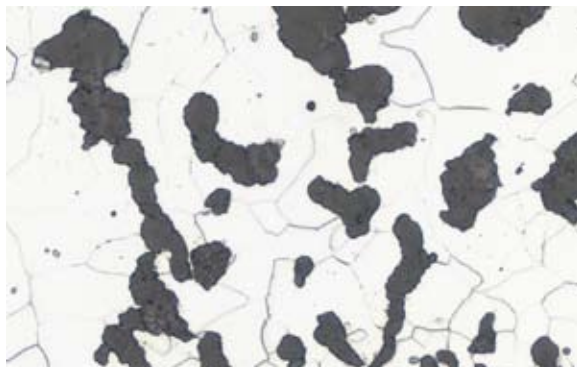
Whether you need to determine the size and shape of graphite particles in cast iron in accordance with EN ISO 945 or the nodularity of vermicular graphite according to SAE J 1887, with this application module you can measure graphite particles fully automatically and with consistent analysis quality.



*Cast iron with lamellar graphite*



*Spheroidal graphite*



*Vermicular graphite*

### Functionality

Developed with practice in mind for the efficient measurement of graphite, the Graphite module analyzes your samples in accordance with the applicable standards. The graphite particles are classified automatically by shape and size. With both methods the results are analyzed according to the specifications of the standards, archived, and documented in appropriate test reports. The Graphite module offers the following methods:

- Determination of shape and size according to EN ISO 945
- Nodularity of vermicular graphite according to SAE J 1887

# Overview

## Functions

| Basic program          |                                  |                                                                                                                              |
|------------------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Image Acquisition      | Function                         | Contents/Description                                                                                                         |
| <b>Image Formats:</b>  | • Image Import                   | zvi, bmp, tif, jpg, j2k, jp2, gif, tga, png, psd, cmp, pct, ras, eps, wmf, mac, msp, img, czi, lsm, vgi, rek, raw, avi, zvhi |
|                        | • Image Export                   | avi, bmp, j2k, jp2, jpg, lsm, mov, pct, pcx, png, psd, tga, tif, wmf                                                         |
| <b>Camera Control:</b> | • Exposure Time Adjustment       | Manual adjustment, exposure time measurement, automatic mode                                                                 |
|                        | • Automatic Exposure Time        | Adjustment of exposure time in live image                                                                                    |
|                        | • Target Value for Exposure Time | Definition of the sensitivity level of the sensor during an exposure measurement                                             |
|                        | • Focus/Exposure Frame           | Optional measurement frame as a focus aid and for spot measurement of the exposure time                                      |
|                        | • Live Image Frame Rate          | Selection: fast/medium/slow for best possible display of the live image                                                      |
|                        | • Resolution                     | Selection of Microscanning resolution modes (AxioCam HR)                                                                     |
|                        | • Binning                        | Increased camera sensitivity by combining the signals of adjacent pixels                                                     |
|                        | • Color Adjustment               | Manual adjustment of the color balance                                                                                       |
|                        | • Color Saturation               | Adjustment of the level of color saturation                                                                                  |
|                        | • Frame                          | Interactive selection of an image sensor sub frame                                                                           |
|                        | • White Balance                  | Interactive or automatic adjustment of optimum neutral balance of the color channels                                         |
|                        | • 3200K                          | Default value for white balance, optimized for halogen light source at 3200 K                                                |
|                        | • Gray Value Scaling             | Adjustment of dynamic range (retain original, convert to 8 bit, convert to 16 bit)                                           |
|                        | • Histogram                      | Intensity distribution histogram for all three color channels                                                                |
|                        | • Black Reference                | Generation of correction image for long exposure times (dark current compensation)                                           |
|                        | • Shading Correction             | Generation of correction image to compensate for optical inhomogeneities                                                     |
|                        | • Image Orientation              | Rotation and mirroring of image orientation for optimum image display                                                        |
|                        | • B/W or Color Mode              | Conversion of color images into monochrome images during acquisition                                                         |
|                        | • Digital Gain                   | Adjustment of digital signal amplification                                                                                   |
|                        | • Analog Gain                    | Analog signal amplification prior to digitization                                                                            |
|                        | • NIR Mode for B/W Cameras       | Mode for further increased sensitivity in near IR for monochrome AxioCam cameras                                             |
|                        | • EMCCD Gain                     | Adjustment of signal amplification for cameras with EMCCD sensor                                                             |
|                        | • CCD Port                       | Selection of amplifier port for cameras with several read-out amplifiers                                                     |
|                        | • Offset                         | Adjustment of basic brightness value                                                                                         |
|                        | • Mode                           | Selection of different, manufacturer-dependent special modes                                                                 |
|                        | • Unsharp Masking                | Sharpening of images immediately during acquisition                                                                          |
|                        | • Trigger Input                  | Triggering of acquisition by means of TTL signal                                                                             |
|                        | • Trigger Output                 | Triggering of a trigger signal, e.g. to control an external shutter                                                          |
| Image Processing       |                                  |                                                                                                                              |
| <b>Annotate:</b>       | • Annotation                     | Addition of text, marking of elements (arrows, scale bars, etc.)                                                             |
| <b>Adjust:</b>         | • Brightness/Contrast/Gamma      | Adjustment of brightness, contrast and gamma                                                                                 |
|                        | • Color Balance                  | Manual adjustment and readjustment of color rendition                                                                        |
|                        | • Hue/Lightness/Saturation       | Adjustment of hue and saturation                                                                                             |
|                        | • Shading Correction             | Correction of uneven illumination                                                                                            |
|                        | • Z-Stack Correction             | Correction of bleaching effects in Z-stack fluorescence images                                                               |
|                        | • Transfer display attributes    | Transfer of display settings (brightness, contrast, gamma) to other images                                                   |
|                        | • Adjust display attributes      | Adjustment of display settings (brightness, contrast, gamma) to pre-defined values                                           |
|                        | • White Balance                  | Change of white balance in a color image                                                                                     |

# Overview

## Functions

|                                  |                        |                                                                                                                      |
|----------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Geometric Transformation:</b> | • Shift                | xyz shift of images                                                                                                  |
|                                  | • Rotate 90            | Rotation of an image by 90°                                                                                          |
|                                  | • Z-Stack Alignment    | Alignment of the individual planes of a Z-stack image which has been acquired, for example, using a stereomicroscope |
|                                  | • OrthoView            | Generation of projections along orthogonal axes in 3D images                                                         |
| <b>Image Smoothing:</b>          | • Gauss, Sigma         | Image smoothing using Gauss or Sigma filter                                                                          |
| <b>Image Sharpening:</b>         | • Enhance Contour      | Enhancement of image sharpness through the intensification of contours                                               |
|                                  | • Unsharp Masking      | Intensification of image sharpness through the enhancement of contrast for small structures and edges                |
| <b>Utilities:</b>                | • Resample             | Reduce/enlarge the size of an image                                                                                  |
|                                  | • Copy Image           | Copy an image and image information that can be selected                                                             |
|                                  | • Load Look-up table   | Load a pseudo-color table                                                                                            |
|                                  | • Export Image         | Export image into other formats                                                                                      |
|                                  | • Convert Pixel Format | Change the pixel format of an image                                                                                  |
|                                  | • Create Image Subset  | Generate a subset from a multi-dimensional image                                                                     |
|                                  | • Add Channels         | Combine images with the same dimensions (Z-stack, time lapse) into multichannel images                               |
|                                  | • Reset Indices        | Re-indexing of image dimensions in ascending order                                                                   |

|                                                      |                            |                                                                                                                                                                                                     |
|------------------------------------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Image Analysis</b>                                |                            |                                                                                                                                                                                                     |
| <b>Interactive Measurement Tools and Parameters:</b> | • Magnetic Cursor          | The cursor detects edges, making it easier to find them, e.g. when measuring lengths                                                                                                                |
|                                                      | • Scalings                 | Scaling in geometric units                                                                                                                                                                          |
|                                                      | • Automatic Scaling        | Automatic detection of pixel size                                                                                                                                                                   |
|                                                      | • Create/Append Table      | Generation/attachment of a data table based on the measurement tools drawn in                                                                                                                       |
|                                                      | • Length, Line             | Definition using 2 points                                                                                                                                                                           |
|                                                      | • Outline/Outline (Spline) | Measurement of diameter, area, perimeter, length and width of the circumscribing rectangle, radius, center of gravity, mean density of gray value, standard deviation of mean density of gray value |
|                                                      | • Angle 3, Angle 4         | Definition using 3 or 4 points                                                                                                                                                                      |
|                                                      | • Circle                   | Measurement of diameter, area, perimeter, length and width of the circumscribing rectangle, radius, center of gravity, mean density of gray value, standard deviation of mean density of gray value |
|                                                      | • Events                   | Counting of events                                                                                                                                                                                  |
|                                                      | • Profile                  | Gray value profile along a line                                                                                                                                                                     |
|                                                      | • Evaluate                 | Functions for the processing and statistical analysis of data tables                                                                                                                                |

|                      |                           |                                                                                                                                |
|----------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Documentation</b> |                           |                                                                                                                                |
|                      | • Gallery                 | Clear presentation of loaded images as thumbnails                                                                              |
|                      | • Info View               | Display of all information for the image                                                                                       |
|                      | • Cut View                | Display of Z-stack images in 3 orthogonal section views (x,y - x,z - y,z)                                                      |
|                      | • Gallery View            | Clear presentation of multidimensional images                                                                                  |
|                      | • Splitter Display        | Comparison of up to 12 images, also multidimensional; generation of comparison as new image document for presentation purposes |
|                      | • Printing of Images/Data | Print of images                                                                                                                |
|                      | • Reports                 | Creation of user-definable reports                                                                                             |

|                      |                              |                                                                    |
|----------------------|------------------------------|--------------------------------------------------------------------|
| <b>My AxioVision</b> |                              |                                                                    |
|                      | • Toolbars/Dialogs/Workflows | Creation of individual toolbars, dialogs and workflows             |
|                      | • Shortcut                   | Allocation of AxioVision functions to keyboard combinations        |
|                      | • Icons                      | Allocation of symbols to AxioVision functions                      |
|                      | • Microscope                 | Allocation of AxioVision functions to up to 10 microscope softkeys |

# Overview

## Functions

| Image Acquisition modules |                                                            |                                                                                                                                                    |
|---------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Z-Stack                   | Acquisition of image series from different focus positions |                                                                                                                                                    |
|                           | • Focus Control                                            | Automatic adjustment of the minimum possible step size according to microscope type                                                                |
|                           | • Z-stack Configuration                                    | Definition of start and stop position (or center position) and interval between individual z-planes                                                |
|                           | • Nyquist Criterion                                        | Automatic calculation of the optimal z-interval                                                                                                    |
|                           | • Navigation                                               | Precise stepwise navigation through defined Z-stack or to the start, stop or center position                                                       |
|                           | • Experiment                                               | Saving of Z-stack definitions as experiment for exact reproduction of an experimental set up                                                       |
|                           | • ReUse                                                    | Extraction of Z-stack definitions from previously acquired images for the exact reproduction of an experimental set up                             |
| Time Lapse                | Flexible acquisition of image series over time             |                                                                                                                                                    |
|                           | • Time Configuration                                       | Definition of interval as well as number of cycles or total time                                                                                   |
|                           | • Exposure Time                                            | Automatic measurement of the correct exposure time for the first time point                                                                        |
|                           | • Image Information                                        | Acquisition time point as annotation in image                                                                                                      |
|                           | • Autosave                                                 | High data security during long time lapse acquisitions thanks to Autosave function                                                                 |
|                           | • Image Size                                               | Acquisition of images as large as required depending on experimental conditions ( > 2 GB)                                                          |
|                           | • Time Lapse Processing                                    |                                                                                                                                                    |
|                           | - Gliding Average                                          | Calculation of average values from time lapse images                                                                                               |
|                           | - Time Differential                                        | Calculation of first and second derivative from time lapse images                                                                                  |
|                           | - Time Concatenate                                         | Combination of two time lapse images to form a new time lapse image                                                                                |
|                           | - Image Ratio                                              | Division of two time lapse images                                                                                                                  |
|                           | - Time Lapse Alignment                                     | Alignment of the individual time points of a time lapse image                                                                                      |
|                           | - Time Stitching                                           | Stitching of heterogeneous ZVI time lapse images to generate one contiguous sequence to enable movie creation from Smart Experiment results        |
|                           | • Experiment                                               | Saving of time lapse configurations as experiment for exact reproduction of an experimental set up                                                 |
|                           | • ReUse                                                    | Extraction of time lapse settings from previously acquired images for the exact reproduction of an experimental set up                             |
|                           | • Smart Experiments                                        | Free combination of different types of experiments to create a Smart Experiment which can be used to acquire heterogeneous multidimensional images |
| Mark&Find                 | Recording and relocating positions                         |                                                                                                                                                    |
|                           | • Database                                                 | Management of projects involving different types of slides in a database (slides, multiple specimen holders, Petri dishes, multiwell plates)       |
|                           | • Mark Interactively                                       | Color assignment of sample positions in the database                                                                                               |
|                           | • Classify                                                 | Assignment of colors and allocation of names for object positions                                                                                  |
|                           | • Select                                                   | Activation/deactivation of individual positions                                                                                                    |
|                           | • Visualize                                                | Visualization of the selected points on a graphic slide. Relocation by clicking on the colored marker                                              |
|                           | • Focus Position                                           | Repositioning with optional use of stored focus position                                                                                           |
|                           | • Import/Export                                            | Import and export of position lists in a file format compatible with Microsoft® Excel                                                              |
|                           | • Calibrate                                                | Calibration using a HOME Calibration slide                                                                                                         |
| MosaiX                    | Automatic scanning of large surfaces                       |                                                                                                                                                    |
|                           | • Execute                                                  | Scanning of the entire surface of a sample (motorized stage required)                                                                              |
|                           | • Focus Correction                                         | Correction of the focus position in the case of uneven sample                                                                                      |
|                           | • Stitching                                                | Correct alignment of tiles to each other                                                                                                           |
|                           | • Convert                                                  | Conversion of tile images into a composite image                                                                                                   |
|                           | • Combinability                                            | MosaiX can be freely combined with all multidimensional image acquisition modules                                                                  |

# Overview

## Functions

|                                     |                                                                                        |                                                                                                                                                                                |
|-------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extended Focus                      | <b>Calculation of sharp images from several focus positions</b>                        |                                                                                                                                                                                |
|                                     | • Acquisition/Computation from Z-stack                                                 | Generation, directly by the camera or from an acquired Z-stack, of an image with extended depth of focus from single images acquired from different focus positions            |
|                                     | • Alignment                                                                            | Correction of the alignment of single images during acquisition with a stereomicroscope                                                                                        |
| Autofocus                           | <b>Automatic focusing</b>                                                              |                                                                                                                                                                                |
|                                     | • Methods                                                                              | Choice between autofocus with calibration and parameter options and autofocus that is always calibrated and does not require parameterization                                  |
|                                     | • Calibrate                                                                            | Calibration by specifying the optimum focus position using the current microscope setting with motorized microscopes                                                           |
|                                     | • Focus                                                                                | Automatic calculation of the optimum focus plane at the touch of a button. Suitable for transmitted-light, reflected-light as well as brightfield, darkfield, and fluorescence |
| Panorama                            | <b>Formation of overview images</b>                                                    |                                                                                                                                                                                |
|                                     | • Acquisition                                                                          | Generation from individually acquired camera images                                                                                                                            |
|                                     | • Import from Files                                                                    | Generation from images that have been saved previously                                                                                                                         |
|                                     | • Stitching                                                                            | Correct alignment of tiles to each other                                                                                                                                       |
|                                     | • Convert                                                                              | Conversion of tile images into a composite image                                                                                                                               |
| HDR Imaging<br>(High Dynamic Range) | <b>Acquisition method for extending the available dynamic range of digital cameras</b> |                                                                                                                                                                                |
|                                     | • HDR Snap                                                                             | Generation and processing of an HDR image using pre-set parameters                                                                                                             |
|                                     | • HDR Series                                                                           | Generation of an HDR raw data image using different exposure times                                                                                                             |
|                                     | • HDR Merge                                                                            | Processing of an HDR raw data image to create an HDR image with offset correction                                                                                              |
|                                     | • HDR Setup                                                                            | Basic setting for activating HDR acquisition for all imaging techniques                                                                                                        |
| <b>Image Processing modules</b>     |                                                                                        |                                                                                                                                                                                |
| Inside4D                            | <b>Visualization in 3D</b>                                                             |                                                                                                                                                                                |
|                                     | • Volume Display                                                                       | Volume display of Z-stack images with up to 8 channels with selective switching between different channels or view in merged pseudo-color mode                                 |
|                                     | • Shadow Projection                                                                    | Creation of animations with strong sense for spatial conditions                                                                                                                |
|                                     | • Transparency Rendering                                                               | Presentation of transparent structures                                                                                                                                         |
|                                     | • Surface Rendering                                                                    | Enhancement of individual structures                                                                                                                                           |
|                                     | • Maximum Projection                                                                   | Ideal for prints and publication                                                                                                                                               |
|                                     | • Mixed Mode                                                                           | Simultaneous display of surface and transparency-rendered data. Simplifies display of small objects within the context of larger structures                                    |
|                                     | • Spatial Interaction                                                                  | Free positioning of the 3D volume in space (with free choice of angles for x, y and z; lateral position and zoom factor)                                                       |
|                                     | • 3D Inside View                                                                       | Orientation within a volume                                                                                                                                                    |
|                                     | • Annotations                                                                          | Optional display of volume edges, color coding and scaling of axes                                                                                                             |
|                                     | • Animations                                                                           | Generation of animations as rendered image series with export options in popular video formats (AVI, QuickTime)                                                                |
|                                     | • Maximum Rendering Speed                                                              | Acceleration of rendering methods through modern graphic boards (support of OpenGL-standard)                                                                                   |
|                                     | • Clipping Planes                                                                      | Exposure of interesting structures by means of up to three freely movable and configurable clipping planes                                                                     |

# Overview

## Functions

| Imaging Plus | Image Improvement, Gray Morphology, Fourier Transformation, Color Transformation |                                                                                                |
|--------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
|              | • Adjust                                                                         |                                                                                                |
|              | - Contrast                                                                       | Contrast enhancement using interactive/automatic histogram adaptation                          |
|              | - Negative                                                                       | Calculation of inverted image (negative)                                                       |
|              | - Gray Transformation                                                            | Adjustment of gray values using transformation tables                                          |
|              | • Geometric Transformations                                                      |                                                                                                |
|              | - Rotate                                                                         | Rotation around an axis                                                                        |
|              | - Mirror                                                                         | Mirror along horizontal or vertical axis                                                       |
|              | - Alignment                                                                      | Affine transformation                                                                          |
|              | - Elastic Registration / Warping                                                 | Alignment using a reference image                                                              |
|              | • Smoothing                                                                      |                                                                                                |
|              | - Denoising                                                                      | Denoising using wavelet transformation                                                         |
|              | - Lowpass                                                                        | Lowpass filter (gliding average)                                                               |
|              | - Median                                                                         | Median filter (non-linear method)                                                              |
|              | - Rank                                                                           | General rank operator                                                                          |
|              | - Gauss Anisotropic                                                              | Anisotropic Gauss filter with selectable Sigma values                                          |
|              | • Sharpening                                                                     |                                                                                                |
|              | - Edge Enhancement                                                               | Enhancement of edges                                                                           |
|              | • Edges                                                                          |                                                                                                |
|              | - Sobel                                                                          | Edge detection using Sobel filter                                                              |
|              | - Laplace                                                                        | Laplace filter                                                                                 |
|              | - Highpass                                                                       | Highpass filter                                                                                |
|              | - Local Variance                                                                 | Edge detection filter calculating local variance of each pixel in relation to its neighborhood |
|              | • Morphology                                                                     |                                                                                                |
|              | - Gray Erode, Gray Dilate                                                        | Erosion or dilation of objects                                                                 |
|              | - Gray Open, Gray Close                                                          | Erosion followed by dilation or dilation followed by erosion                                   |
|              | - Tophat White                                                                   | Removal of bright regions                                                                      |
|              | - Tophat Black                                                                   | Accentuation of dark regions                                                                   |
|              | - Gray Gradient                                                                  | Morphological gradient to detect contours                                                      |
|              | - Watersheds                                                                     | Watersheds – algorithm for separation/reconstruction                                           |
|              | • Arithmetics                                                                    |                                                                                                |
|              | - Add, Subtract                                                                  | Addition or subtraction of two images                                                          |
|              | - Add Constant                                                                   | Addition of a constant value                                                                   |
|              | - Multiply, Divide                                                               | Multiplication or division of two images                                                       |
|              | - Multiply Constant                                                              | Multiplication with a constant value                                                           |
|              | - Average                                                                        | Average of two images                                                                          |
|              | - Maximum, Minimum                                                               | Maximum or minimum of two images                                                               |
|              | - Square, SquareRoot                                                             | Square or square root of an image                                                              |
|              | - Logarithm, Exponential                                                         | Logarithm or exponent of an image                                                              |
|              | - Combine                                                                        | Linear combination of two images                                                               |
|              | • FFT                                                                            |                                                                                                |
|              | - Transform                                                                      | Fourier transformation on an image                                                             |
|              | - Spectrum                                                                       | Calculation of power or phase spectrum                                                         |
|              | - Filter                                                                         | Filtering in the frequency domain using a defined filter                                       |
|              | - Inverse                                                                        | Inverse Fourier transformation                                                                 |

# Overview

## Functions

|                           |                                                                                                                                             |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| • Utilities               |                                                                                                                                             |
| - Copy Region             | Copying of image regions                                                                                                                    |
| - Color Model             | Transformation of RGB color space into HLS color space and vice versa                                                                       |
| - Split RGB Extractions   | Splitting of RGB image into single color channels                                                                                           |
| - Combine RGB Extractions | Combination of single color channels to form a color image                                                                                  |
| - User Filter             | Filtering of an image with user-defined filter matrix                                                                                       |
| - Generate Noise          | Superimposing an image with predefined noise                                                                                                |
| • Time Lapse Processing   |                                                                                                                                             |
| - Gliding Average         | Calculation of average values from time lapse images                                                                                        |
| - Time Differential       | Calculation of first and second derivative from time lapse images                                                                           |
| - Time Concatenate        | Combination of two time lapse images to create a new time lapse image                                                                       |
| - Image Ratio             | Division of two time lapse images                                                                                                           |
| - Time Lapse Alignment    | Alignment of the individual time points of a time lapse image                                                                               |
| - Time Stitching          | Stitching of heterogeneous ZVI time lapse images to generate one contiguous sequence to enable movie creation from Smart Experiment results |

| Image Analysis modules                   |                                                                                         |
|------------------------------------------|-----------------------------------------------------------------------------------------|
| Interactive Measurement                  | Expanded interactive measurement techniques                                             |
| • Distance, Calipers                     | Measurement of length                                                                   |
| • Multiple Calipers/Distance             | Measurement of the length of multiple lines, perpendicular to a base line               |
| • Curve, Curve (Spline)                  | Measurement of length of the drawn curve                                                |
| • Aligned rectangle or free orientation  | Measurement of geometric and densitometric object features                              |
| • Circle (Radius), Circle (Points)       | Drawing of a radius to the center, clicking on contour points                           |
| • Marker                                 | x and y coordinates of a point                                                          |
| • Points, Relative Points                | x and y coordinates of one or more points with free definition of the coordinate system |
| • Interactive Measurement Program Wizard | Guided generation of a program for interactive measurement                              |
| • Interactive Measurement Programs       | Loading and execution of interactive measurement programs                               |

| Online Measurement            | Interactive measurements in online images                            |
|-------------------------------|----------------------------------------------------------------------|
| • Activate Online Measurement | Execution of interactive measurements in an online image             |
| • Layer                       | Predefined and individual grids can be displayed in the online image |

| AutoMeasure                      | Creation of easy measurement programs with a measurement wizard |                                                                                                                                                        |
|----------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Creation of Measurement Programs | • Automatic Measurement Program Wizard                          | Guided generation of a program for automatic measurement                                                                                               |
|                                  | • Image Enhancement                                             | Contrast, brightness, Gamma, noise reduction (Sigma), shading correction, improvement of edges                                                         |
|                                  | • Segmentation                                                  | Global or local definition by clicking or circumscribing objects, specification of thresholds using the image histogram, definition of multiple phases |
|                                  | • Binary Image Clean-up                                         | Deletion of artifacts, filling of holes                                                                                                                |
|                                  | • Automatic Object Separation                                   | Erosion and dilation, watersheds                                                                                                                       |
|                                  | • Editing of the Measurement Mask                               | Drawing of separation lines, deletion of objects, addition of objects                                                                                  |
|                                  | • Selection of Measurement Parameters                           | Region-specific, field-specific, geometric and annotation parameters, user-defined parameters                                                          |
|                                  | • Definition of Measurement Conditions ("objectfilter")         | Logical concatenation (and/or) of region-specific parameters, definition by simple clicking of reference objects                                       |
|                                  | • Definition of a Measurement Frame                             | Rectangle, circle, freehand                                                                                                                            |
|                                  | • Measurement                                                   | Measurement of geometric and densitometric features for single objects or the entire image                                                             |
|                                  | • Documentation                                                 | Marking of measured objects and display of freely selectable measurement parameters in the graphics plane                                              |
|                                  | • Data Storage                                                  | Saving of measurement data in a Microsoft® Excel-compatible file format (CSV, XML)                                                                     |

# Overview

## Functions

|                                          |                       |                                                                                                          |
|------------------------------------------|-----------------------|----------------------------------------------------------------------------------------------------------|
| <b>Execution of Measurement Programs</b> | • Image Acquisition   | Image acquisition via camera, all images of a folder, all loaded images                                  |
|                                          | • Control of Program  | Activation/deactivation as well as the changing of functional parameters during execution of the program |
|                                          | • Program Information | List of executed functions with parameter settings                                                       |

|                         |                                                                     |                                                                                                                                             |
|-------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AutoMeasure Plus</b> | <b>Segmentation, binary image processing, automatic measurement</b> |                                                                                                                                             |
|                         | • Segmentation                                                      |                                                                                                                                             |
|                         | - Thresholds                                                        | Interactive adjustment of thresholds with histogram support and specification of fixed values                                               |
|                         | - Region Growing                                                    | Detection of associated regions (gray values within user-defined tolerance range)                                                           |
|                         | - Multiphase                                                        | Adjustment of thresholds for several phases of an image with histogram support                                                              |
|                         | - Automatic                                                         | Automatic determination of thresholds using a histogram                                                                                     |
|                         | - Dynamic                                                           | Technique for threshold detection using size information                                                                                    |
|                         | - Valleys                                                           | Detection of dark lines (valleys) in images with bright background                                                                          |
|                         | - Canny                                                             | Edge detection considering "steepness" of edges                                                                                             |
|                         | - Marr                                                              | Detection of edges and associated regions                                                                                                   |
|                         | • Binary Functions                                                  |                                                                                                                                             |
|                         | - Erode, Dilate                                                     | Erosion or dilation of binary objects                                                                                                       |
|                         | - Ultimate Erode                                                    | Erosion of binary objects while keeping the smallest structures                                                                             |
|                         | - Open, Close                                                       | Erosion followed by dilation or dilation followed by erosion                                                                                |
|                         | - Clean Up Binary Image                                             | Filling of holes, removal of artifacts                                                                                                      |
|                         | - Mark regions                                                      | Marking of regions using a mask image                                                                                                       |
|                         | - Object Separation                                                 | Automatic separation of touching regions                                                                                                    |
|                         | - Binary Image Editor                                               | Interactive subsequent editing (separating, combining) of binary images                                                                     |
|                         | - AND, OR, XOR, NOT                                                 | Bit-by-bit "logic" operations                                                                                                               |
|                         | - Distance Transformation                                           | Generation of a "distance map", indicating the distance of each pixel to the object border                                                  |
|                         | • Skeletonizing of Binary Images                                    |                                                                                                                                             |
|                         | - Thinning                                                          | Thinning of binary objects to lines 1 pixel wide ("skeleton")                                                                               |
|                         | - Skeleton                                                          | Skeletonization of the image background                                                                                                     |
|                         | • Selection of Measurement Parameters                               | Region-specific, field-specific, geometric and annotation parameters, user-defined parameters                                               |
|                         | - Definition of Measurement Conditions ("objectfilter")             | Logical concatenation (and, or) of region-specific parameters, definition by simple clicking of reference objects                           |
|                         | - Definition of a Measurement Frame                                 | Rectangle, circle, freehand                                                                                                                 |
|                         | - Measurement                                                       | Automatic measurement of geometric and densitometric object features, drawing in of measurement values into the graphics plane of the image |

|                       |                                                                   |                                                                                                                                             |
|-----------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>3D Measurement</b> | <b>Measurement of three-dimensional structures and parameters</b> |                                                                                                                                             |
|                       | • Interactive Measurement in 3D Space                             | Drawing in of lines, angles, markers and curves in rendered 3D views                                                                        |
|                       | • Segmentation                                                    | Interactive adjustment of thresholds in rendered 3D view and with specification of fixed values                                             |
|                       | • Binary Image Editor                                             | Interactive subsequent editing (separating, combining) of 3D binary images                                                                  |
|                       | • Measurement                                                     | Automatic measurement of geometric and densitometric object features, drawing in of measurement values into the graphics plane of the image |

# Overview

## Functions

| Particle Analyzer Projects          | Measurement of particles                                                                                       |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------|
| • System solution                   | Coordinated components: microscope, camera, PC, software                                                       |
| • Automatic classification          | Implemented standards, e.g. VDA 19, ISO 16232, user-defined adjustment of standards                            |
| - Particle classification           | Reflective particles, non-reflective particles, fibers                                                         |
| • User modes                        |                                                                                                                |
| - Administrator                     | Definition of the measurement procedure using test specifications                                              |
| - User                              | Execution of the measurement                                                                                   |
| • Data management                   | Based on Asset Archive, selection of the data to be saved                                                      |
| • Results display                   |                                                                                                                |
| - Results view                      | Display of the classification results for each standard and method                                             |
| - Gallery of the largest inclusions | View the largest inclusions, relocate the inclusions under the microscope, remove artifacts, various galleries |
| • Report                            | Issue of standard-compliant reports, modification possible                                                     |

| NMI                                      | Determination of the content of non-metallic inclusions in rolled steel                                        |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| • System solution                        | Coordinated components: microscope, camera, PC, software                                                       |
| • Multiple samples                       | Acquisition and analysis of several samples in a single measurement                                            |
| • Simultaneous analysis of all standards | EN 10247, DIN 50602, ASTM E 45, JIS G 0555, ISO 4967                                                           |
| • User modes                             |                                                                                                                |
| - Administrator                          | Definition of the measurement procedure using test specifications                                              |
| - User                                   | Execution of the measurement                                                                                   |
| • Data management                        | Based on Asset Archive, selection of the data to be saved                                                      |
| • Results display                        |                                                                                                                |
| - Results view                           | Display of the classification results for each standard and method                                             |
| - Gallery of the largest inclusions      | View the largest inclusions, relocate the inclusions under the microscope, remove artifacts, various galleries |
| • Report                                 | Issue of standard-compliant reports, modification possible                                                     |

| Grains              | Determination of grain size                                                                                                                     |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| • Measurement modes |                                                                                                                                                 |
| - Interactive       | Comparison using comparative diagrams in accordance with ASTM – Plate I, Plate II, Plate III, Plate IV, SEP 151061, BS 4990                     |
| - Chord             | Intercept method in accordance with DIN EN 623, ASTM E 112, ASTM E 1382, various line patterns                                                  |
| - Automatic         | Automatic reconstruction of grain boundaries in accordance with DIN EN 623, ASTM E 112, ASTM E 1382, additional measurement parameters possible |
| • User modes        |                                                                                                                                                 |
| - Administrator     | Definition of the measurement procedure using test specifications                                                                               |
| - User              | Execution of the measurement                                                                                                                    |
| • Data management   | Based on Asset Archive                                                                                                                          |
| • Report            | Issue of standard-compliant reports, modification possible                                                                                      |

| Multiphase      | Analysis of multiphase samples                                                                            |
|-----------------|-----------------------------------------------------------------------------------------------------------|
| • Measurement   | Phase components as a percentage, classification of particle sizes, free choice of measurement parameters |
| • User modes    |                                                                                                           |
| - Administrator | Definition of the measurement procedure using test specifications                                         |
| - User          | Execution of the measurement                                                                              |

# Overview

## Functions

|                   |                                         |
|-------------------|-----------------------------------------|
| • Data management | Based on Asset Archive                  |
| • Report          | Issue of reports, modification possible |

| Graphite | Analysis of graphite particles in cast iron |                                                                                |
|----------|---------------------------------------------|--------------------------------------------------------------------------------|
|          | • Measurement modes                         |                                                                                |
|          | - Shape and size                            | Determination of shape and size in accordance with EN ISO 945                  |
|          | - SinterCast                                | Determination of nodularity in accordance with SAE J 1887                      |
|          | - Flake Iron                                | Determination of the size of lamellar graphite in accordance with EN ISO 945   |
|          | - Gray Iron                                 | Determination of the size of spheroidal graphite in accordance with EN ISO 945 |
|          | • User modes                                |                                                                                |
|          | - Administrator                             | Definition of the measurement procedure using test specifications              |
|          | - User                                      | Execution of the measurement                                                   |
|          | • Data management                           | Based on Asset Archive                                                         |
|          | • Report                                    | Issue of standard-compliant reports, modification possible                     |

| Comparative Diagrams | Interactive comparison of comparative diagrams with micrographs |                                                                                |
|----------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------|
|                      | • Display                                                       | Comparison with one, two, four or eight comparative diagrams                   |
|                      | • Wizard                                                        | Creation of user-defined comparative diagrams                                  |
|                      | • User modes                                                    |                                                                                |
|                      | - Administrator                                                 | Definition of the measurement procedure using test specifications              |
|                      | - User                                                          | Execution of the measurement                                                   |
|                      | • Data management                                               | Based on Asset Archive                                                         |
|                      | • Report                                                        | Display of images, data tables and statistical analysis, modification possible |

| Layer Thickness | Measurement of layer thickness |                                                                   |
|-----------------|--------------------------------|-------------------------------------------------------------------|
|                 | Layer detection                | Gray or color-value segmentation or interactive detection         |
|                 | Measurement                    | Individual or several layers, straight or curved layers           |
|                 | • User modes                   |                                                                   |
|                 | - Administrator                | Definition of the measurement procedure using test specifications |
|                 | - User                         | Execution of the measurement                                      |
|                 | • Data management              | Based on Asset Archive                                            |
|                 | • Report                       | Issue of results, modification possible                           |

| Calotte Grinding | Measurement of layers in accordance with DIN V ENV 1071 (calotte grinding method) |                                                                                                         |
|------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
|                  | • Drawing in of measurement circles                                               | Drawing in of measurement circles for single or multilayer                                              |
|                  | • Analysis                                                                        | Automatic calculation of layer thickness, display of measurement results in image, generation of report |

| TIC | Optical height measurement with Total Interference Contrast |                                                              |
|-----|-------------------------------------------------------------|--------------------------------------------------------------|
|     | • TIC settings                                              | Correction of phase shift and objective aperture             |
|     | • Automatic TIC measurement                                 | Measurement of fringe displacement in the interference image |

# Overview

## Functions

|                                                |                                                            |                                                                                                                   |
|------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Topography</b>                              | <b>Height and roughness measurement in 3D topographies</b> |                                                                                                                   |
|                                                | • Generation of topographies                               |                                                                                                                   |
|                                                | - Calculation from stereo image pairs                      | Calculation of the topographic image from a stereoscopic image pair                                               |
|                                                | - Calculation from Z-stack images                          | Generation of the topographic image from a Z-stack by means of surface recognition                                |
|                                                | - Composition from texture image and height map            | Generation of the topographic image from a texture image and a specified height map                               |
|                                                | • Topographic views                                        |                                                                                                                   |
|                                                | - Texture                                                  | Display of the structure of the detected surface                                                                  |
|                                                | - Height image                                             | Display of the height map in gray-value or pseudo-color coding                                                    |
|                                                | - 3D projections                                           | 3D display of the topography in gray-value or pseudo-color coding, texture projection, grid or surface projection |
|                                                | - Anaglyph                                                 | Display of the topography as a stereogram that can be viewed using anaglyph glasses                               |
|                                                | • Measurements                                             |                                                                                                                   |
|                                                | - Profile measurement                                      | Measurement and display of height profiles of lines, polygons and curves                                          |
|                                                | - Roughness measurement in accordance with EN ISO 4287     | Calculation of roughness statistics from primary, roughness and waviness profiles                                 |
|                                                | - Cut-off wavelength                                       | Setting of the cut-off wavelength for measuring roughness and waviness                                            |
|                                                | - Flood height                                             | Setting of a flood height for displaying regions of equal height                                                  |
|                                                | • Generation of views                                      | Each topographic view can be saved as an image                                                                    |
|                                                | • Generation of measurement value lists                    |                                                                                                                   |
|                                                | - Height profiles                                          | Height profiles can be saved as a graph or a data list                                                            |
|                                                | - Roughness statistics                                     | The roughness statistics of the drawn-in profile lines can be saved as a data list                                |
| <b>Documentation and Configuration modules</b> |                                                            |                                                                                                                   |
| <b>Asset Archive</b>                           | <b>Archiving of images, measurement data, and reports</b>  |                                                                                                                   |
|                                                | • Structured Archiving of Assets                           | Allocation of assets to projects, contacts, and categories                                                        |
|                                                | • Search                                                   | Keyword search and freely definable search queries on the basis of field content                                  |
|                                                | • Value Lists                                              | Data entry using adaptable value lists                                                                            |
|                                                | • Local Management of Archives                             | Single-user system, storage location for the database may be selected                                             |
| <b>Commander</b>                               | <b>Recording/execution of steps</b>                        |                                                                                                                   |
|                                                | • Record, Save                                             | Recording of work steps and saving of scripts                                                                     |
|                                                | • Start                                                    | Automatic execution of recorded scripts                                                                           |
|                                                | • Edit                                                     | Subsequent editing of scripts                                                                                     |
| <b>VBA</b>                                     | <b>Integrated development environment</b>                  |                                                                                                                   |
|                                                | • Visual Basic Editor                                      | VBA environment with full access to AxioVision functionalities                                                    |

# Overview

## Region-specific Measurement Parameters

| Region-specific parameters |                    |                         |             |            |                                                  |                                                                                                                                 |
|----------------------------|--------------------|-------------------------|-------------|------------|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| • Geometric parameters     |                    |                         |             |            |                                                  |                                                                                                                                 |
| Entry-level                | Materials Packages | Interactive Measurement | AutoMeasure | 3D Measure |                                                  |                                                                                                                                 |
|                            |                    | X                       | X           | X          | AcpX, AcpY                                       | x and y coordinates of the first object point of a region                                                                       |
|                            |                    |                         |             | X          | AcpZ                                             | z coordinate of the first object point of a 3D region                                                                           |
| X                          | X                  | X                       | X           |            | Area                                             | Area of the region in scaled and unscaled units                                                                                 |
|                            |                    | X                       | X           |            | Area convex, Area filled                         | Area of the convex shell of the region and of the filled region                                                                 |
|                            |                    |                         | X           |            | Area to area sum                                 | Area of the region in relation to the total area of all regions                                                                 |
|                            |                    |                         | X           |            | Area to Frame area                               | Area of the region in relation to the area of the measurement frame                                                             |
|                            |                    |                         |             | X          | Surface, SurfaceFilled                           | Surface content of the 3D region and of the filled 3D region                                                                    |
|                            |                    |                         |             | X          | Volume                                           | Volume of the 3D region in scaled and unscaled units                                                                            |
|                            |                    |                         |             | X          | Volume filled                                    | Volume of the filled 3D region                                                                                                  |
|                            |                    |                         |             | X          | Volume to volume sum                             | Volume of the 3D region in relation to the total volume of all 3D regions                                                       |
|                            |                    |                         |             | X          | Volume to Frame Volume                           | Volume of the 3D region in relation to the volume of the measurement frame                                                      |
|                            |                    |                         | X           | X          | Count of inner parts                             | Number of holes and regions within holes                                                                                        |
| X                          | X                  | X                       | X           | X          | CenterX, CenterY                                 | x and y coordinates of the geometric center of gravity of the region                                                            |
|                            |                    |                         |             | X          | CenterZ                                          | z coordinate of the geometric center of gravity of the 3D region                                                                |
|                            |                    | X                       | X           | X          | Ellipse major, Ellipse minor                     | Length of the main axis and the secondary axis of the ellipse with the same geometric moment of inertia as the region/3D region |
|                            |                    |                         |             | X          | Ellipse Semi-Medial Axis                         | Length of the middle axis of the ellipse with the same geometric moment of inertia as the 3D region                             |
|                            |                    | X                       | X           | X          | Ellipse angle                                    | Angle of the main axis of the ellipse with the same moment of inertia                                                           |
| X                          | X                  | X                       | X           |            | Perimeter                                        | Perimeter of the region                                                                                                         |
|                            |                    | X                       | X           |            | Perimeter convex                                 | Perimeter of the convex shell of the region                                                                                     |
|                            |                    | X                       | X           |            | Perimeter filled                                 | Perimeter of the filled region                                                                                                  |
|                            |                    | X                       | X           |            | Perimeter Crofton, Perimeter Crofton filled      | Perimeter of the region and perimeter of the filled region according to Crofton                                                 |
|                            |                    | X                       | X           |            | Perimeter X, Perimeter Y                         | x and y projection of the perimeter                                                                                             |
|                            |                    | X                       | X           |            | Perimeter XF, Perimeter YF                       | x and y projection of the perimeter of the filled region                                                                        |
|                            |                    | X                       | X           |            | Perimeter XY, Perimeter XYF                      | Diagonal projection of the perimeter and the perimeter of the filled region                                                     |
|                            | X                  | X                       | X           | X          | Bound left, Bound top, Bound right, Bound bottom | x and y coordinates of the bounding box/the bounding cuboid of a 3D region                                                      |
|                            |                    |                         |             | X          | Bound front, Bound back                          | z coordinates of the bounding cuboid of a 3D region                                                                             |
| X                          | X                  | X                       | X           | X          | Bound width, Bound height                        | Width and height of the bounding box/the bounding cuboid of a 3D region                                                         |
|                            |                    |                         |             | X          | Bound depth                                      | Depth of the bounding cuboid of a 3D region                                                                                     |
|                            | X                  |                         | X           |            | Area Frame                                       | Area of the measurement frame in scaled and unscaled units                                                                      |
|                            |                    |                         |             | X          | Volume Frame                                     | Volume of the measurement frame in scaled and unscaled units                                                                    |
|                            | X                  | X                       | X           | X          | Feret minimum, Feret maximum                     | Minimum and maximum feret of the region                                                                                         |

# Overview

## Region-specific and Field-specific Measurement Parameters

| Region-specific parameters |                    |                         |             |            |                                            |                                                                                      |
|----------------------------|--------------------|-------------------------|-------------|------------|--------------------------------------------|--------------------------------------------------------------------------------------|
| • Geometric parameters     |                    |                         |             |            |                                            |                                                                                      |
| Entry-level                | Materials Packages | Interactive Measurement | AutoMeasure | 3D Measure |                                            |                                                                                      |
|                            |                    | X                       | X           |            | Feret Min. Angle, Feret Max. Angle         | Angle of the minimum and the maximum feret of the region                             |
|                            |                    |                         |             | X          | Feret Min. Azimut, Feret Max. Azimut       | Horizontal orientation of the minimum and the maximum feret of the 3D region         |
|                            |                    |                         |             | X          | Feret Min. Elevation, Feret Max. Elevation | Vertical orientation of the minimum and the maximum feret of the 3D region           |
|                            | X                  | X                       | X           | X          | Feretratio                                 | Ratio of the ferets (Feret Min/Feret Max)                                            |
| X                          | X                  | X                       | X           | X          | Diameter, Radius                           | Diameter, radius of the circle with equivalent area/sphere with equivalent volume    |
|                            | X                  | X                       | X           | X          | Form circle, Form sphere                   | Circular shape factor of the region/spherical shape factor of the 3D region          |
|                            |                    | X                       | X           |            | Fibrelength                                | Length of a fiber-like thin region                                                   |
|                            | X                  | X                       | X           |            | Index/ID                                   | Explicit characteristic of the region, of the squares                                |
| X                          | X                  | X                       |             | X          | Distance, Length                           | Distance between 2 points, length of a line                                          |
|                            | X                  | X                       |             |            | Distances Mean                             | Mean distance of multiple distances                                                  |
| X                          | X                  | X                       |             | X          | Angle Measurement                          | Angle in degrees                                                                     |
| • Densitometric parameters |                    |                         |             |            |                                            |                                                                                      |
| X                          | X                  | X                       |             | X          | Mean                                       | Densitometric mean value of the region (gray and color values)                       |
| X                          | X                  | X                       | X           | X          | Standard Deviation                         | Standard deviation of the densitometric values of the region (gray and color values) |
|                            |                    | X                       | X           | X          | Minimum, Maximum                           | Minimum and maximum densitometric value (gray and color values)                      |
|                            |                    | X                       | X           | X          | Sum                                        | Sum of the densitometric values of the region                                        |
|                            |                    | X                       |             | X          | Sum Square                                 | Sum of the squares (gray and color values)                                           |
| Field-specific parameters  |                    |                         |             |            |                                            |                                                                                      |
| • Geometric parameters     |                    |                         |             |            |                                            |                                                                                      |
|                            |                    |                         | X           |            | Area sum                                   | Area of all regions in scaled and unscaled units                                     |
|                            |                    |                         | X           |            | Area sum filled                            | Area of all filled regions                                                           |
|                            |                    |                         | X           |            | Area percent                               | Percentage area of all regions in the measurement frame                              |
|                            |                    |                         | X           | X          | Number of regions                          | Number of the measured regions                                                       |
|                            |                    |                         | X           |            | Perimeter sum                              | Sum of all region perimeters                                                         |
|                            |                    |                         |             | X          | Surface sum                                | Surface content of all 3D regions                                                    |
|                            |                    |                         |             | X          | Volume sum                                 | Volume of all 3D regions in scaled and unscaled units                                |
|                            |                    |                         |             | X          | Volume sum filled                          | Volume of all filled 3D regions                                                      |
|                            |                    |                         |             | X          | Volume percent                             | Percentage volume of all 3D regions in the measurement cuboid                        |

# Overview

## Field-specific and Image-specific Measurement Parameters

| Field-specific parameters                                                                                                                                                                                              |                    |                         |             |            |                           |                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------------|-------------|------------|---------------------------|--------------------------------------------------------------------------------|
| • Densitometric parameters                                                                                                                                                                                             |                    |                         |             |            |                           |                                                                                |
| Entry-level                                                                                                                                                                                                            | Materials Packages | Interactive Measurement | AutoMeasure | 3D Measure |                           |                                                                                |
|                                                                                                                                                                                                                        |                    |                         | X           | X          | Mean                      | Densitometric mean value of all regions (gray and color values)                |
|                                                                                                                                                                                                                        |                    |                         | X           | X          | Standard Deviation        | Densitometric value standard deviation in all regions (gray and color values)  |
|                                                                                                                                                                                                                        |                    |                         | X           | X          | Minimum, Maximum          | Minimum and maximum densitometric value in all regions (gray and color values) |
| Further parameters                                                                                                                                                                                                     |                    |                         |             |            |                           |                                                                                |
| X                                                                                                                                                                                                                      | X                  | X                       |             | X          | Count                     | Number of objects clicked on                                                   |
|                                                                                                                                                                                                                        |                    | X                       |             | X          | Marker                    | x and y coordinates of an object                                               |
| X                                                                                                                                                                                                                      | X                  | X                       |             |            | Gray/Color Value Profiles | Gray value/color value along a profile line                                    |
|                                                                                                                                                                                                                        |                    |                         | X           | X          | User Parameter            | Parameter that can be defined by the user                                      |
| Image-specific parameters                                                                                                                                                                                              |                    |                         |             |            |                           |                                                                                |
| X                                                                                                                                                                                                                      | X                  | X                       | X           | X          | Name                      | Name of the image                                                              |
| X                                                                                                                                                                                                                      | X                  | X                       | X           | X          | Acquisition Time          | Time point at which the image was acquired                                     |
| X                                                                                                                                                                                                                      | X                  | X                       | X           | X          | Exposure Time             | Exposure time of the image                                                     |
| X                                                                                                                                                                                                                      | X                  | X                       | X           | X          | Focus Position            | Focus position of the image                                                    |
| X                                                                                                                                                                                                                      | X                  | X                       | X           | X          | Microscope Magnification  | Microscope magnification set during image acquisition                          |
| X                                                                                                                                                                                                                      | X                  | X                       | X           | X          | Date Saved                | Date on which the acquired image was saved                                     |
| X                                                                                                                                                                                                                      | X                  | X                       | X           | X          | Stage Position X, Y       | x and y stage position at which the image was acquired                         |
| X                                                                                                                                                                                                                      |                    | X                       | X           | X          | Channel Name              | Name of the channel for multichannel images                                    |
| X                                                                                                                                                                                                                      |                    | X                       | X           | X          | Phase Name/Index          | Phase name/index for multiphase images                                         |
| X                                                                                                                                                                                                                      |                    | X                       | X           | X          | Index/ID Channel          | Index/ID of the channel of the multichannel image                              |
| X                                                                                                                                                                                                                      |                    | X                       | X           | X          | Index/ID Z-plane          | Z-index/ID for Z-stack images                                                  |
| X                                                                                                                                                                                                                      |                    | X                       | X           | X          | Index/ID Time             | Time index/ID for time lapse images                                            |
| Statistical parameters                                                                                                                                                                                                 |                    |                         |             |            |                           |                                                                                |
| Minimum, Maximum, Mean, Count, Sum, Standard Deviation, Range, Sum of squares, Variance, 25-Quartile, 50-Quartile (Median), 75-Quartile, 10-Percentile, 90-Percentile, 1-Percentile, 99-Percentile, Kurtosis, Skewness |                    |                         |             |            |                           |                                                                                |



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